Find the propagation velocities and the directions of the phenomena described by the following functions.

a) $u(x,t) = sin(2x - 3t)$

b) $u(x,t) = cos(4x + 6t)$

c) $u(z,t) = sin\left(\frac{3}{2}z - 3t\right) cos\left(\frac{1}{2}z - 9t\right)$

d) $u(x,t) = sin[2(x + 3t)]$

e) $u(x,t) = sin(2x)cos(6t)$

f) $u(y,t) = sin^2(y + 3t)$

g) $u(x,t) = x^2 \cos(6t)$: Left to the students.

h) $u(x,t) = 2tcos(3x)$: Left to the students.

Solution:

a) $u(x,t) = sin(2x - 3t) = sin\left(2\left(x - \frac{3}{2}t\right)\right) = f(x - ct)$ where $f(\xi) = \sin(2\xi)$.

This is a wave phenomenon which propagates in the $+x$ direction with velocity $c = \frac{3}{2}$ m/s.

b) $u(x,t) = cos(4x + 6t) = cos\left(4\left(x + \frac{6}{4}t\right)\right) = cos\left(4\left(x + \frac{3}{2}t\right)\right) = g(x + ct)$ where $g(\xi) = \cos(4\xi)$.

This is a wave phenomenon which propagates in the $-x$ direction with velocity $c = \frac{3}{2}$ m/s.

c) $u(z,t) = sin\left(\frac{3}{2}z - 3t\right) cos\left(\frac{1}{2}z - 9t\right) = \frac{1}{2}sin\left(\frac{3}{2}z - 3t + \frac{1}{2}z - 9t\right) + \frac{1}{2}sin\left(\frac{3}{2}z - 3t - \frac{1}{2}z + 9t\right)$

$$u(z,t) = \frac{1}{2}sin\big(2(z - 6t)\big) + \frac{1}{2}sin(z + 6t) = f(z - ct) + g(z + ct)$$

where $f(\xi) = \frac{1}{2}\sin(2\xi)$ and $g(\xi) = \frac{1}{2}\sin(\xi)$. This is a wave phenomenon which propagates in the $-z$ and $+z$ direction with velocity $c = 6$ m/s.

Hint: $sin(A) \times cos(B) = \frac{1}{2}sin(A + B) + \frac{1}{2}sin(A - B)$

d) $u(x,t) = sin[2(4x + 6t)] = sin\left[8\left(x + \frac{3}{2}t\right)\right] = g(x + ct)$ where $g(\xi) = \sin(8\xi)$

This is a wave phenomenon which propagates in the $-x$ direction with velocity $c = \frac{3}{2}$ m/s.

e) $u(x,t) = sin(2x)cos(6t) = \frac{1}{2}\sin(2x + 6t) + \frac{1}{2}\sin(2x - 6t)$

$$u(x,t) = sin(2x)cos(6t) = \frac{1}{2}\sin[2(x - 3t)] + \frac{1}{2}\sin[2(x + 3t)] = f(x - ct) + g(x + ct)$$

This is a wave phenomenon propagating in both $-x$ and $+x$ directions with velocity $c = 3$ m/s.

f) $u(y,t) = sin^2(y + 3t) = g\,(y + ct)$ where $g(\xi) = sin^2(\xi)$.

This is a wave phenomenon which propagates in the $-y$ direction with velocity $c = 3$ m/s.

Consider an electromagnetic wave propagates in a lossless simple medium of infinite extent $x \in (-\infty, \infty)$ with the time interval of $t \in [0, \infty)$. It satisfies the wave equation as

$$\frac{\partial^2}{\partial x^2}u(x,t) - 4\frac{\partial^2}{\partial t^2}u(x,t) = 0$$

under the initial conditions $u(x,0) = 0$ and $\frac{\partial u(x,0)}{\partial t} = e^{-|x|}$.

a) What is the velocity of the wave?
b) Find the expression of $u(x,t)$ at the time $t = x$ and in the half space $x > 0$.
c) Find the expression of $u(x,t)$ at the time $t = 4x$ and in the half space $x > 0$.

Solution

$$u_0(x) = u(x,0) = 0 \quad , \quad v_0(x) = \frac{\partial u(x,0)}{\partial t} = e^{-|x|}$$

a)

$$\frac{\partial^2}{\partial x^2}u(x,t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}u(x,t) = \frac{\partial^2}{\partial x^2}u(x,t) - 4\frac{\partial^2}{\partial t^2}u(x,t) \quad \Rightarrow \quad \frac{1}{c^2} = 4 \quad \Rightarrow \quad c = \sqrt{\frac{1}{4}} \quad \boxed{= \frac{1}{2}\ \text{m/s}}$$

c can not have negative values ($2c = 1\ m/s$).

b) Using d'Alembert solution of the wave equation:

$$u(x,t) = \frac{1}{2}\{u_0(x-ct) + u_0(x+ct)\} + \frac{1}{2c}\int_{x-ct}^{x+ct} v_0(\gamma)d\gamma$$

$$u_0(x) = u(x,0) = 0 \quad \Rightarrow \quad u_0(x-ct) = u_0(x+ct) = 0 \quad \Rightarrow \quad u(x,t) = \frac{1}{2c}\int_{x-ct}^{x+ct} v_0(\gamma)d\gamma$$

$$v_0(\gamma) = e^{-|\gamma|} \quad \Rightarrow \quad u(x,t) = \frac{1}{2c}\int_{x-\frac{1}{2}t}^{x+\frac{1}{2}t} e^{-|\gamma|}d\gamma = \frac{\overbrace{1}^{\frac{1}{2}}}{2\frac{1}{2}}\int_{x-\frac{1}{2}t}^{x+\frac{1}{2}t} e^{-|\gamma|}d\gamma$$

$$t = x \quad \Rightarrow \quad u(x,t)|_{t=x} = u(x,x) = \int_{x-\frac{1}{2}x}^{x+\frac{1}{2}x} e^{-|\gamma|}d\gamma = \int_{\frac{1}{2}x}^{\frac{3}{2}x} e^{-\gamma}d\gamma = -e^{-\gamma}|_{\frac{1}{2}x}^{\frac{3}{2}x} = \boxed{e^{-\frac{1}{2}x} - e^{-\frac{3}{2}x}}$$

c)

$$t = 4x \quad \Rightarrow \quad u(x,t)|_{t=4x} = u(x,4x) = \int_{x-\frac{1}{2}4x}^{x+\frac{1}{2}4x} e^{-|\gamma|}d\gamma = \int_{x-2x}^{x+2x} e^{-|\gamma|}d\gamma = \int_{-x}^{3x} e^{-|\gamma|}d\gamma = \underbrace{\int_{-x}^{0} e^{+\gamma}d\gamma}_{e^{\gamma}|_{-x}^{0}} + \underbrace{\int_{0}^{3x} e^{-\gamma}d\gamma}_{-e^{-\gamma}|_{0}^{3x}}$$

$$u(x,t)|_{t=4x} = (1 - e^{-x}) + (-e^{-3x} + 1) = \boxed{2 - e^{-x} - e^{-3x}}$$

Consider an electromagnetic wave which propagates in a lossless simple medium of infinite extent $x \in (-\infty, \infty)$, $t \in [0, \infty)$ and satisfies the wave equation at below

$$\frac{\partial^2}{\partial x^2}u(x,t) - 16\frac{\partial^2}{\partial t^2}u(x,t) = 0$$

$$u(x,0) = 0$$

$$\frac{\partial u(x,0)}{\partial t} = sin\,(x)$$

a) What is the **propagation velocity** of the wave?
b) Find the **solution** $u(x,t)$ of the above-given initial-value problem.

Solution:
a)

$$u_0(x) = u(x,0) = 0 \quad , \quad v_0(x) = \frac{\partial u(x,0)}{\partial t} = sin\,(x)$$

$$\frac{\partial^2}{\partial x^2}u(x,t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}u(x,t) = \frac{\partial^2}{\partial x^2}u(x,t) - 16\frac{\partial^2}{\partial t^2}u(x,t) \quad \Rightarrow \quad \frac{1}{c^2} = 16 \quad \Rightarrow \quad c = \sqrt{\frac{1}{16}} = \frac{1}{4}\ m/s$$

c can not have negative values.

b)

Using d'Alembert solution of the wave equation:

$$u(x,t) = \frac{1}{2}\{u_0(x-ct) + u_0(x+ct)\} + \frac{1}{2c}\int_{x-ct}^{x+ct} v_0(\gamma)d\gamma$$

$$u_0(x) = u(x,0) = 0 \quad \Rightarrow \quad u_0(x-ct) = u_0(x+ct) = 0 \quad \Rightarrow \quad u(x,t) = \frac{1}{2c}\int_{x-ct}^{x+ct} v_0(\gamma)d\gamma$$

$$u(x,t) = \frac{1}{2c}\int_{x-\frac{1}{4}t}^{x+\frac{1}{4}t} sin\,(\gamma)d\gamma = \frac{\overbrace{1}^{2}}{2\frac{1}{4}}\left[-cos(\gamma)\Big|_{x-\frac{1}{4}t}^{x+\frac{1}{4}t}\right]$$

$$\Rightarrow \quad u(x,t) = 2\left[cos\left(x-\frac{1}{4}t\right) - cos\left(x+\frac{1}{4}t\right)\right]$$

The initial conditions of an electromagnetic wave propagating in a source-free and lossless simple medium bounded by planes $x = 0$ and $x = a$ are given as

$$u(x,0) = 0 \quad , \quad u_t(x,0) = cos\left(\frac{\pi}{a}x\right)$$

and the boundary conditions of these planes

$$u_x(0,t) = 0 \quad , \quad u_x(a,t) = 0.$$

Find the explicit expression of the wave $u(x,t)$ for all $x\epsilon[0,a]$ and $t \geq 0$.

Solution:

$$u_{xx} - \frac{1}{c^2}u_{tt} = 0 \quad , \quad x\epsilon[0,a] \quad ; \quad t \geq 0$$

$$u_x(0,t) = 0 \quad , \quad u_x(a,t) = 0 \; ; \; Neumann\ boundary\ condition$$

The solution of this wave equation should be in the form of (due to similarity of $u_t(x,0)$ given above)

$$u(x,t) = \sum_{n=0}^{\infty} a_n(t) cos\left(\frac{n\pi}{a}x\right)$$

substituting it into the wave equation

$$u_{xx} - \frac{1}{c^2}u_{tt} = 0 \quad \Rightarrow \quad \sum_{n=0}^{\infty}\left[-\frac{n^2\pi^2}{a^2}a_n(t) - \frac{1}{c^2}a_n''(t)\right]cos\left(\frac{n\pi}{a}x\right) = 0$$

$$\Rightarrow \quad -\frac{n^2\pi^2}{a^2}a_n(t) - \frac{1}{c^2}a_n''(t) = 0 \quad \Rightarrow \quad \boxed{a_n''(t) + \left(\frac{n\pi c}{a}\right)^2 a_n(t) = 0}$$

where proposing a solution for the differential equation of $a_n(t)$

$$a_n(t) = A_n cos\left(\frac{n\pi c}{a}t\right) + B_n sin\left(\frac{n\pi c}{a}t\right)$$

applying the first initial conditions $(u(x,0) = 0)$

$$u(x,0) = \sum_{n=0}^{\infty} a_n(0) cos\left(\frac{n\pi}{a}x\right) = 0 \quad \Rightarrow \quad a_n(0) = 0 \quad \Rightarrow \quad \boxed{A_n = 0}$$

and applying the second initial conditions $(u_t(x,0) = cos\left(\frac{\pi}{a}x\right))$

$$u_t(x,0) = \sum_{n=0}^{\infty} a_n'(0) cos\left(\frac{n\pi}{a}x\right) = cos\left(\frac{\pi}{a}x\right) \quad \Rightarrow \quad \sum_{n=0}^{\infty}\frac{n\pi c}{a}B_n cos\left(\frac{n\pi}{a}x\right) = cos\left(\frac{\pi}{a}x\right)$$

then equating them

$$\frac{\pi c}{a}B_1 cos\left(\frac{\pi}{a}x\right) + \frac{2\pi c}{a}B_2 cos\left(\frac{2\pi}{a}x\right) + \cdots = cos\left(\frac{\pi}{a}x\right) \quad \Rightarrow \quad \left\{\begin{array}{c} B_2 = B_3 = \cdots = 0 \\ \frac{\pi c}{a}B_1 = 1 \;\Rightarrow\; \boxed{B_1 = \frac{a}{\pi c}} \end{array}\right\} \quad \Rightarrow \quad B_n = \begin{cases} \frac{a}{\pi c} & ; \; n = 1 \\ 0 & ; \; n \neq 1 \end{cases}$$

$$u(x,t) = \sum_{n=0}^{\infty} B_n sin\left(\frac{n\pi c}{a}t\right) cos\left(\frac{n\pi}{a}x\right) = B_1 sin\left(\frac{\pi c}{a}t\right) cos\left(\frac{\pi}{a}x\right)$$

$$\boxed{u(x,t) = \frac{a}{\pi c} sin\left(\frac{\pi c}{a}t\right) cos\left(\frac{\pi}{a}x\right)}$$

Consider a wave phenomenon represented by

$$u(x,t) = cos(2x + 6\pi.10^6 t) + 2cos(2x - 6\pi.10^6 t)$$

a) What are the physical interpretations (propagation directions, velocities, frequencies, wavelengths and amplitudes) of the first and second terms?
b) Observe that $u(x,t)$ can also be written as

$$u(x,t) = 2cos(2x)\cos(6\pi.10^6 t) + cos(2x - 6\pi.10^6 t)$$

Give the physical interpretation of the first and second terms appearing in this new expression.

Solution

a)

$$u(x,t) = \underbrace{cos(2x + 6\pi.10^6 t)}_{u_1(x,t)} + \underbrace{2cos(2x - 6\pi.10^6 t)}_{u_2(x,t)}$$

• $u_1(x,t) = cos(2x + 6\pi.10^6 t) = cos[2(x + 3\pi.10^6 t)] = cos(-2x - 6\pi.10^6 t)$

The propagation of direction: $-x$

The velocity: $c_1 = 3\pi.10^6 \text{ m/s} = \frac{\omega}{|\nabla\alpha|} = \frac{6\pi.10^6}{|-2\hat{x}|} = \frac{6\pi.10^6}{2} = 3\pi.10^6 \text{ m/s} = v_{phase_1}$

The frequency: $f_1 = \frac{\omega_1}{2\pi} = \frac{6\pi.10^6}{2\pi} = 3.10^6 \text{ Hz} = 3 \text{ MHz}$

The wavelength: $\lambda_1 = v_{phase_1} T_1 = v_{phase_1} \frac{1}{f_1} = \frac{3\pi.10^6\ m/s}{3.10^6\ Hz} = \pi \text{ m}$

The amplitude: $A_1 = 1$

• $u_2(x,t) = 2cos(2x - 6\pi.10^6 t) = 2cos[2(x - 3\pi.10^6 t)]$

The propagation of direction: $+x$

The velocity: $c_2 = 3\pi.10^6 \text{ m/s} = \frac{\omega}{|\nabla\alpha|} = \frac{6\pi.10^6}{|2\hat{x}|} = \frac{6\pi.10^6}{2} = 3\pi.10^6 \text{ m/s} = v_{phase_2}$

The frequency: $f_2 = \frac{\omega_2}{2\pi} = \frac{6\pi.10^6}{2\pi} = 3.10^6 \text{ Hz} = 3 \text{ MHz}$

The wavelength: $\lambda_2 = v_{phase_2} T_2 = v_{phase_2} \frac{1}{f_2} = \frac{3\pi.10^6\ m/s}{3.10^6\ Hz} = \pi \text{ m}$

The amplitude: $A_2 = 2$

b)

$$u(x,t) = \underbrace{2cos(2x)\cos(6\pi.10^6 t)}_{u_3(x,t)} + \underbrace{cos(2x - 6\pi.10^6 t)}_{u_4(x,t)}$$

• $u_3(x,t) = 2cos(2x)\cos(6\pi.10^6 t) = A(x)cos(\omega t)$

Due to the phase of $u_3(x,t)$ is only time dependent, $u_3(x,t)$ is a *Standing Wave*.

• $u_4(x,t) = cos(2x - 6\pi.10^6 t)$

The propagation of direction: $+x$

The velocity: $c_4 = 3\pi.10^6\ m/s = \frac{\omega}{|\nabla\alpha|} = \frac{6\pi.10^6}{|2\vec{e}_x|} = \frac{6\pi.10^6}{2} = 3\pi.10^6\ m/s = v_{phase_2}$

The frequency: $f_4 = \frac{\omega_4}{2\pi} = \frac{6\pi.10^6}{2\pi} = 3.10^6\ \text{Hz} = 3\ \text{MHz}$

The wavelength: $\lambda_4 = v_{phase_4} T_4 = v_{phase_4} \frac{1}{f_4} = \frac{3\pi.10^6\ m/s}{3.10^6\ Hz} = \pi\ \ m$

The amplitude: $A_4 = 1$

Note: Let's evaluate the field of $u(x,t) = cos(2x + 6\pi.10^6 t) + 2cos(2x - 6\pi.10^6 t)$

$u(x,t) = cos(2x + 6\pi.10^6 t) + cos(2x - 6\pi.10^6 t) + cos(2x - 6\pi.10^6 t)$

$= cos(2x)cos(6\pi.10^6 t) - sin(2x)sin(6\pi.10^6 t) + cos(2x)cos(6\pi.10^6 t) + sin(2x)sin(6\pi.10^6 t)$
$+cos(2x - 6\pi.10^6 t)$

$= cos(2x)cos(6\pi.10^6 t) \underbrace{-sin(2x)sin(6\pi.10^6 t) + sin(2x)sin(6\pi.10^6 t)}_{0} + cos(2x)cos(6\pi.10^6 t)$
$+cos(2x - 6\pi.10^6 t)$

$u(x,t) = cos(2x)cos(6\pi.10^6 t) + cos(2x)cos(6\pi.10^6 t) + cos(2x - 6\pi.10^6 t)$

$u(x,t) = 2cos(2x)cos(6\pi.10^6 t) + cos(2x - 6\pi.10^6 t)$

This shows that a standing wave (amplitude 2) and the $+x$ propagated wave (amplitude 1) are equal to the $-x$ propagated (amplitude 1) and $+x$ propagated wave (amplitude 2). It means that a standing wave (amplitude 2) is equal to the $-x$ propagated (amplitude 1) and $+x$ propagated wave (amplitude 1).

Write the complex expressions of the following monochromatic waves:

a) $u(x,t) = -cos(x-3t)$

b) $\vec{\boldsymbol{H}}(x,y,t) = e^{-x}cos\left(\sqrt{2}x - y + 3.10^6 t\right)\hat{\boldsymbol{z}}$

c) $u(z,t) = \cos(z)\sin(2t)$

Find the time depending real expressions of the following complex fields with $\omega = 3.10^8$ (rad/sec):

d) $U(x) = -icos(3x)$

e) $\vec{\boldsymbol{H}}(x,y) = e^{-x-i(x+y)}\hat{\boldsymbol{z}}$

f) $\vec{\boldsymbol{E}}(z) = -\sin(z)\hat{\boldsymbol{y}}$

Solution:

Complex notation:

a) $u(x,t) = -cos(x-3t) = Re\left\{e^{i(\pi+x-3t)}\right\} = Re\left\{e^{i\pi}e^{ix}e^{-i3t}\right\} \Rightarrow u(x) = -e^{ix}$

b) $\vec{\boldsymbol{H}}(x,y,t) = e^{-x}cos\left(\sqrt{2}x - y + 3.10^6 t\right)\hat{\boldsymbol{z}} = Re\left\{e^{-x}e^{i(-\sqrt{2}x+y-3.10^6 t)}\right\}\hat{\boldsymbol{z}} \Rightarrow \vec{\boldsymbol{H}}(x,y) = e^{-x}e^{i(-\sqrt{2}x+y)}\hat{\boldsymbol{z}}$

c)

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$$u(z,t) = \cos(z)\sin(2t) = \frac{1}{2}sin(z+2t) + \frac{1}{2}sin(2t-z) = \frac{1}{2}\left(cos\left(\frac{\pi}{2} - z - 2t\right) + cos\left(\frac{\pi}{2} + z - 2t\right)\right)$$

$$u(z,t) = \frac{1}{2}Re\left\{e^{i\frac{\pi}{2}}.e^{-iz}.e^{-i2t} + e^{i\frac{\pi}{2}}.e^{+iz}.e^{-i2t}\right\} = Re\left\{\frac{1}{2}i(e^{+iz} + e^{-iz}).e^{-i2t}\right\} = Re\left\{icos(z).e^{-i2t}\right\}$$

$$U(z) = icos(z)$$

Real notation:

d) $u(x,t) = Re\left\{u(x)e^{-i\omega t}\right\} = Re\left\{-i\cos(3x)\ \underbrace{e^{-i3.10^8 t}}_{\cos\omega t - i\sin\omega t}\right\} = -\cos 3x\ \sin(3.10^8 t)$

e) $\vec{\boldsymbol{H}}(x,y,t) = Re\left\{e^{-x}e^{-i(x+y)}e^{-i3.10^8 t}\right\}\hat{\boldsymbol{z}} = e^{-x}\cos(-x-y-3.10^8 t)\hat{\boldsymbol{z}}$

$$\vec{\boldsymbol{H}}(x,y,t) = e^{-x}\cos(x+y+3.10^8 t)\hat{\boldsymbol{z}}$$

f) $\vec{\boldsymbol{E}}(z,t) = Re\left\{-\sin(z)e^{-i.3.10^8 t}\hat{\boldsymbol{y}}\right\} = -\sin(z)\cos(3.10^8 t)\hat{\boldsymbol{y}}$

Write the complex expressions of the following monochromatic waves:

a) $V(x,t) = -2sin(4x-5t)$

b) $\vec{\boldsymbol{E}}(x,y,t) = e^{-x}cos\left(\sqrt{2}x+y-3.10^6t\right)\hat{\boldsymbol{z}}$

c) $u(z,t) = sin(2z)cos(6t)$

Find the time depending real expressions of the following complex fields ($\omega = 2$ rad/sec):

d) $U(x) = isin(3x)$

e) $\vec{\boldsymbol{E}}(x,y) = e^{-3x+i(x+y)}\hat{\boldsymbol{z}}$

f) $\vec{\boldsymbol{H}}(z) = 3cos(z)\hat{\boldsymbol{y}}$

Solution:

Complex notation:

a)

$$V(x,t) = -2sin(4x-5t) = 2sin(-4x+5t) = 2cos\left(\frac{\pi}{2}+4x-5t\right) = Re\left\{2e^{i\left(\frac{\pi}{2}+4x-5t\right)}\right\} = Re\left\{2e^{i\left(\frac{\pi}{2}+4x\right)}e^{-i5t}\right\}$$

$$\Rightarrow \quad V(x) = 2e^{i\left(\frac{\pi}{2}+4x\right)} = 2e^{i\frac{\pi}{2}}e^{i4x} \quad \Rightarrow \quad \boxed{V(x) = i2e^{i4x}}$$

b) $\vec{\boldsymbol{E}}(x,y,t) = e^{-x}cos\left(\sqrt{2}x+y-3.10^6t\right)\hat{\boldsymbol{z}} = Re\left\{e^{-x}e^{i\left(\sqrt{2}x+y\right)}e^{-i3.10^6t}\right\}\hat{\boldsymbol{z}} \quad \Rightarrow \quad \boxed{\vec{\boldsymbol{E}}(x,y) = e^{-x}e^{i\left(\sqrt{2}x+y\right)}}$

c) $u(z,t) = sin(2z)cos(6t) = \frac{1}{2}(sin(2z+6t)+sin(2z-6t)) = \frac{1}{2}\left(cos\left(\frac{\pi}{2}-2z-6t\right)+cos\left(\frac{\pi}{2}-2z+6t\right)\right)$

$$u(z,t) = sin(2z)cos(6t) = \frac{1}{2}cos\left(\frac{\pi}{2}-2z-6t\right)+\frac{1}{2}cos\left(-\frac{\pi}{2}+2z-6t\right)$$

$$u(z,t) = \frac{1}{2}Re\left\{e^{i\left(\frac{\pi}{2}-2z-6t\right)}\right\}+\frac{1}{2}Re\left\{e^{i\left(-\frac{\pi}{2}+2z-6t\right)}\right\} = \frac{1}{2}Re\left\{e^{i\left(\frac{\pi}{2}-2z\right)}e^{-i6t}\right\}+\frac{1}{2}Re\left\{e^{i\left(-\frac{\pi}{2}+2z\right)}e^{-i6t}\right\}$$

$$u(z,t) = \frac{1}{2}Re\left\{\left(e^{i\left(\frac{\pi}{2}-2z\right)}+e^{i\left(-\frac{\pi}{2}+2z\right)}\right)e^{-i6t}\right\} \quad \Rightarrow \quad u(z) = \frac{1}{2}\left(e^{i\left(\frac{\pi}{2}-2z\right)}+e^{i\left(-\frac{\pi}{2}+2z\right)}\right) = \frac{1}{2}ie^{-i2z}-\frac{1}{2}ie^{+i2z}$$

$$u(z) = \frac{1}{2}ie^{-i2z}-\frac{1}{2}ie^{+i2z} = \frac{1}{2}i\left(e^{-i2z}-e^{+i2z}\right) = \frac{1}{2}i(-i2sin(2z)) \quad \Rightarrow \quad \boxed{u(z) = sin(2z)}$$

Real notation:

d) $U(x,t) = Re\{U(x)e^{-i\omega t}\} = Re\left\{isin(3x)\underbrace{e^{-i2t}}_{cos(2t)-i\,sin(2t)}\right\} = Re\{isin(3x)(cos(2t)-i\,sin(2t))\}$

$$U(x,t) = Re\{isin(3x)(cos(2t)-i\,sin(2t))\} = Re\{isin(3x)\,cos(2t)+sin(3x)sin\,(2t)\} = \boxed{sin(3x)sin\,(2t)}$$

e) $\vec{\boldsymbol{E}}(x,y,t) = Re\{\vec{\boldsymbol{E}}(x,y)e^{-i\omega t}\} = Re\{e^{-3x}e^{i(x+y)}\hat{\boldsymbol{z}}e^{-i2t}\} = e^{-3x}\,Re\{e^{i(x+y-2t)}\}\hat{\boldsymbol{z}}$

$$\vec{\boldsymbol{E}}(x,y,t) = e^{-3x}Re\{cos(x+y-2t)+isin(x+y-2t)\}\hat{\boldsymbol{z}} = \boxed{e^{-3x}\cos(x+y-2t)\hat{\boldsymbol{z}}}$$

f) $\vec{\boldsymbol{H}}(z,t) = Re\{\vec{\boldsymbol{H}}(z)e^{-i\omega t}\} = Re\{3cos(z)\hat{\boldsymbol{y}}e^{-i2t}\} = 3\,Re\{cos(z)(cos(2t)-isin(2t))\}\hat{\boldsymbol{y}}$

$$\vec{\boldsymbol{H}}(z,t) = 3\,cos(z)Re\{\cos(2t)-i\sin(2t)\}\hat{\boldsymbol{y}} = \boxed{3\,cos(z)\cos(2t)\,\hat{\boldsymbol{y}}}$$

Suppose that an electromagnetic wave propagating in a lossless simple medium ($\varepsilon_r = 2$ and $\varepsilon_0 = (1/36\pi).10^{-9}$ F/m) not involving the sources has the following magnetic field

$$\vec{H}(x,t) = 3.10^{-6}cos(3x - 9.10^8 t)\vec{y}$$

Find the real expression of the electric field $\vec{E}(x,t)$.

Solution:

$$\vec{H}(x,t) = 3.10^{-6}cos(3x - 9.10^8 t)\vec{y}$$

$$\vec{H}(x,t) = \mathrm{Re}\{3.10^{-6}e^{i3x}e^{-i9.10^8 t}\vec{y}\} = \mathrm{Re}\{\vec{H}(x)e^{-i\omega t}\}$$

$$\vec{H}(x) = H(x)\vec{y} = 3.10^{-6}e^{i3x}\vec{y}$$

where $\omega = 9.10^8$ rad/s.

Now, let us consider the second Maxwell's equation in the complex notation

$$\nabla \times \vec{H}(x) + i\omega\vec{D}(x) = \underbrace{\vec{J}_c(x) + \vec{J}_v(x)}_{=\vec{J}(x)}$$

$$\vec{J}_c(x) = 0 \ \ in\ lossless\ medium\ (\sigma = 0)$$

$$\vec{J}_v(x) = 0 \ \ in\ sourceless\ medium\ (\rho = 0)$$

$$\vec{D}(x) = \varepsilon\vec{E}(x)\ in\ simple\ medium\ (\varepsilon \neq 0)$$

Then

$$\nabla \times \vec{H}(x) + i\omega\varepsilon\vec{E}(x) = 0 \quad \Rightarrow \quad \vec{E}(x) = -\frac{\nabla \times \vec{H}(x)}{i\omega\varepsilon} = \frac{i}{\omega\varepsilon}\nabla \times \vec{H}(x)$$

where $\varepsilon = \varepsilon_r\varepsilon_0$

$$\Rightarrow \quad \vec{E}(x) = \frac{i}{\omega\varepsilon}\nabla \times \vec{H}(x) = \frac{i}{\omega\varepsilon}\begin{vmatrix} \vec{x} & \vec{y} & \vec{z} \\ \frac{\partial}{\partial x} & 0 & 0 \\ 0 & H_y & 0 \end{vmatrix} = \frac{i}{\omega\varepsilon}\left(\frac{\partial}{\partial x}\vec{x}\right) \times (3.10^{-6}e^{i3x}\vec{y}) = \frac{i}{9.10^8\,\varepsilon_r\varepsilon_0}3.10^{-6}i3e^{i3x}\vec{z}$$

$$\Rightarrow \quad \vec{E}(x) = -\frac{10^{-14}}{2\varepsilon_0}e^{i3x}\vec{z} = -18\pi 10^{-5}e^{i3x}\vec{z}$$

The real expression (time-domain)

$$\vec{E}(x,t) = \mathrm{Re}\{\vec{E}(x)e^{-i\omega t}\} = \mathrm{Re}\{-18\pi 10^{-5}e^{i3x}e^{-i9.10^8 t}\vec{z}\}$$

$$\Rightarrow \quad \vec{E}(x,t) = -18\pi 10^{-5}cos\,(3x - 9.10^8 t)\vec{z}$$

Alternatively, it can be represented that

$$\vec{E}(x,t) = -18\pi 10^{-5}sin\left(\frac{\pi}{2} - 3x + 9.10^8 t\right)\vec{z}$$

$$\Rightarrow \quad \vec{E}(x,t) = 18\pi 10^{-5}sin\left(3x - \frac{\pi}{2} - 9.10^8 t\right)\vec{z}.$$

The electric field of a monochromatic wave propagating in a source-free simple medium ($\mu_r = 1, \sigma \neq 0$) is given by

$$\vec{E}(y,t) = 5e^{-y}sin(-3y + 6.10^8 t)\vec{x}$$

a) Find k, ε and σ of the medium.
b) Determine the wavelength and phase velocity.
c) Find the complex and real expressions of magnetic field.
d) Compute the average electric field energy (W_E) stored in the region $|x| \leq 3, |y| \leq 2, |z| \leq 1$.
e) Compute the average magnetic field energy (W_H) as function of W_E stored in the region $|x| \leq 3, |y| \leq 2, |z| \leq 1$.

Solution

a)

$$\vec{E}(y,t) = 5e^{-y}sin(-3y + 6.10^8 t)\vec{x} = 5e^{-y}cos\left(\frac{\pi}{2} + 3y - 6.10^8 t\right)\vec{x}$$

$$\vec{E}(y,t) = \text{Re}\left\{\underbrace{5e^{-y}e^{i\pi/2}e^{i3y}}_{E(y)}\, e^{-i6.10^8 t}\vec{x}\right\} \quad ; \quad \omega = 6.10^8 \text{ rad/s}$$

$$\vec{E}(y) = E(y)\vec{x} = 5ie^{-y}e^{i3y}\vec{x}$$

Now, let us use Helmholtz equation

$$\Delta\vec{E} + k^2\vec{E} = 0 \Rightarrow \frac{\partial^2\vec{E}}{\partial y^2} + k^2\vec{E} = 0 \Rightarrow \frac{\partial^2}{\partial y^2}\left(5ie^{-y}e^{i3y}\vec{x}\right) + k^2\left(5ie^{-y}e^{i3y}\vec{x}\right) = 0$$

$$\Rightarrow [(-1+i3)^2 + k^2]\vec{E} = 0 \Rightarrow (-1+i3)^2 + k^2 = 0 \Rightarrow k^2 = -(-1+i3)^2 = -1 + 9 + i6 = 8 + i6$$

$$\Rightarrow k^2 = 10\left(\frac{4}{5} + i\frac{3}{5}\right) = 10\left(cos(\theta) + isin(\theta)\right) = 10e^{i\theta} \Rightarrow k = \sqrt{10}e^{i\theta/2}$$

$$\Rightarrow k = \sqrt{10}\left(cos\left(\frac{\theta}{2}\right) + isin\left(\frac{\theta}{2}\right)\right)$$

To find θ, let us use a trigonometric relation

$$cos\theta = 2\cos^2\frac{\theta}{2} - 1 = \frac{4}{5} \Rightarrow cos\frac{\theta}{2} = \frac{3}{\sqrt{10}} \Rightarrow sin\frac{\theta}{2} = \frac{1}{\sqrt{10}}$$

$$\Rightarrow k = \sqrt{10}\left(\frac{3}{\sqrt{10}} + i\frac{1}{\sqrt{10}}\right) = \boxed{3 + i}$$

To calculate ε and σ, let us use k

$$k^2 = \omega^2\varepsilon\mu + i\sigma\omega\mu = 8 + i6 \Rightarrow \begin{cases} \omega^2\varepsilon\mu = 8 \\ \sigma\omega\mu = 6 \end{cases}$$

$$\omega^2 \underbrace{\varepsilon_r\varepsilon_0}_{\varepsilon}\underbrace{\mu_r\mu_0}_{\mu} = 8 \Rightarrow \varepsilon_r = \frac{8}{\omega^2\varepsilon_0\mu_r\mu_0} = \frac{8c_0^2}{\omega^2\mu_r} = \frac{8.(3.10^8)^2}{(6.10^8)^2.1} = \frac{72}{36} \Rightarrow \boxed{\varepsilon_r = 2}$$

$$\sigma\omega\underbrace{\mu_r\mu_0}_{\mu} = 6 \Rightarrow \sigma = \frac{6}{\omega\mu_r\mu_0} = \frac{6}{\omega\mu_0} = \frac{6}{6.10^8 \times 4\pi.10^{-7}} = \frac{1}{40\pi} \Rightarrow \boxed{\sigma = \frac{1}{40\pi}\ \text{S/m}}$$

b)

$$\alpha = \frac{\pi}{2} + 3y \quad \Rightarrow \quad \lambda = v_{phase}T = \frac{\omega}{|\nabla\alpha|}\cdot\frac{1}{f} = \frac{2\pi}{|\nabla\alpha|} = \frac{2\pi}{\left|\nabla\left(\frac{\pi}{2}+3y\right)\right|} = \frac{2\pi}{|3\vec{y}|} = \boxed{\frac{2\pi}{3}\ \text{m}}$$

$$v_{phase} = \frac{\omega}{|\nabla\alpha|} = \frac{6.10^8}{3} = \boxed{2.10^8\ \text{m/s}}$$

c)

$$\nabla\times\vec{E} - i\omega\mu_0\vec{H} = 0 \quad \Rightarrow \quad \vec{H} = -i\frac{\nabla\times\vec{E}}{\omega\mu_0} = -\frac{i}{\omega\mu_0}\begin{vmatrix}\vec{x} & \vec{y} & \vec{z}\\ 0 & \frac{\partial}{\partial y} & 0\\ E_x & 0 & 0\end{vmatrix} = \frac{i}{\omega\mu_0}\left(\frac{\partial}{\partial y}E_x\right)\vec{z}$$

$$\vec{H} = \frac{i}{\omega\mu_0}5i(-1+i3)e^{-y}e^{i3y}\vec{z} = \frac{5(1-i3)}{240\pi}e^{-y}e^{i3y}\vec{z} = \frac{5(1-i3)}{240\pi}e^{y(-1+i3)}\vec{z}$$

$$\vec{H}(y,t) = \text{Re}\{\vec{H}(y)e^{-i\omega t}\} = \text{Re}\left\{\frac{5(1-i3)}{240\pi}e^{y(-1+i3)}\vec{z}e^{-i\omega t}\right\} = \frac{5}{240\pi}\text{Re}\{(1-i3)e^{y(-1+i3)}e^{-i\omega t}\}\vec{z}$$

$$\vec{H}(y,t) = \frac{5}{240\pi}\text{Re}\{e^{y(-1+i3)}e^{-i\omega t} - i3e^{y(-1+i3)}e^{-i\omega t}\}\vec{z} = \frac{5}{240\pi}\text{Re}\{e^{-y}e^{i(3y-\omega t)} - i3e^{-y}e^{i(3y-\omega t)}\}\vec{z}$$

$$= \frac{5e^{-y}}{240\pi}\text{Re}\{\cos(3y-\omega t) + i\sin(3y-\omega t) - i3\cos(3y-\omega t) + 3\sin(3y-\omega t)\}\vec{z}$$

$$= \frac{5e^{-y}}{240\pi}\text{Re}\{[\cos(3y-\omega t) + 3\sin(3y-\omega t)] + i[\sin(3y-\omega t) - 3\cos(3y-\omega t)]\}\vec{z}$$

$$= \frac{5e^{-y}}{240\pi}[\cos(3y-\omega t) + 3\sin(3y-\omega t)]\vec{z} = \boxed{\frac{e^{-y}}{48\pi}[\cos(3y-6.10^8t) + 3\sin(3y-6.10^8t)]\vec{z}}$$

d)

$$W_E = \frac{\varepsilon}{4}\int_v |\vec{E}|^2 dv = \frac{2\varepsilon_0}{4}\int_{x=-3}^{3}\int_{y=-2}^{2}\int_{z=-1}^{1} 25\,e^{-2y}dxdydz = \frac{25\varepsilon_0}{2}\times 2\times 6\int_{y=-2}^{2} e^{-2y}dy$$

$$W_E = -\frac{25\varepsilon_0}{2}\times 6e^{-2y}\Big|_{-2}^{+2} = \boxed{75\varepsilon_0(e^4 - e^{-4})\ \text{Joule}}$$

e)

$$W_H = \frac{\mu}{4}\int_v |\vec{H}|^2 dv = \frac{\mu_0}{4}\int_{x=-3}^{3}\int_{y=-2}^{2}\int_{z=-1}^{1}\frac{25}{(240)^2\pi^2}10e^{-2y}dxdydz = \frac{25\mu_0}{4\times(24)^2100\pi^2}10\times 6\times 2\times\left(-\frac{1}{2}\right)(e^{-4}-e^4)$$

$$W_H = \frac{75\times 4\pi.10^{-7}}{2\times(24)^2 10\pi^2}(e^4 - e^{-4}) = \frac{75\times 10^{-7}}{2\times 24\times 6\times 10\pi}(e^4-e^{-4}) = 75\frac{10}{8}\frac{\overbrace{10^{-9}}^{\varepsilon_0}}{36\pi}(e^4-e^{-4}) = \frac{5}{4}\underbrace{75\varepsilon_0(e^4-e^{-4})}_{W_E}$$

$$\Rightarrow \quad \boxed{W_H = \frac{5}{4}W_E}$$

where it means that the reactive power is not zero.

An electromagnetic plane wave propagating in a non-magnetic source free simple medium is observed as

$$\vec{H}(y,z,t) = -5sin\left(2y + 2\sqrt{3}z - 3.10^8 t\right)\vec{x}$$

a) Find the complex expression of the $\vec{H}(y,z)$.
b) By using the result obtained in **a)**, write the wave number k and the propagation direction $\vec{n}$.
c) Determine the conductivity σ, relative permittivity ε_r and as well as the characteristic impedance Z of the medium.
d) Compute the complex expression of the $\vec{E}(y,z)$ by using a relation between $\vec{E}$, $\vec{H}$ and $\vec{n}$.
e) Show that $\vec{E}$, $\vec{H}$ and $\vec{n}$ are perpendicular to each other.

Solution

w nin başı eksi ve cos olacak

a)

$$\vec{H}(y,z,t) = -5sin\left(2y + 2\sqrt{3}z - 3.10^8 t\right)\vec{x} = +5sin\left(-2y - 2\sqrt{3}z + 3.10^8 t\right)\vec{x}$$

$\cos(0) = 1$, $\sin\left(\frac{\pi}{2}\right) = 1$, $\sin(x) = \cos\left(x - \frac{\pi}{2}\right)$

$$\vec{H}(y,z,t) = +5cos\left(\frac{\pi}{2} + 2y + 2\sqrt{3}z - 3.10^8 t\right)\vec{x}$$

$$\vec{H}(y,z,t) = \mathrm{Re}\left\{+5e^{i\left(\frac{\pi}{2}+2y+2\sqrt{3}z-3.10^8 t\right)}\right\}\vec{x} = \mathrm{Re}\left\{+5e^{i\left(\frac{\pi}{2}+2y+2\sqrt{3}z\right)}e^{-i3.10^8 t}\right\}\vec{x}$$

$$\vec{H}(y,z) = +5e^{i\left(\frac{\pi}{2}+2y+2\sqrt{3}z\right)}\vec{x} = +5\underbrace{e^{i\left(\frac{\pi}{2}\right)}}_{i}e^{i(2y+2\sqrt{3}z)}\vec{x} \quad \Rightarrow \quad \boxed{\vec{H}(y,z) = +5ie^{i2(y+\sqrt{3}z)}\vec{x}}$$

b) The wave is a plane wave ($\omega = 3.10^8$ rad/s). Therefore, $\vec{H}(y,z)$ can be written in a form of

$$\vec{H}(y,z) = \vec{H}_0 e^{i\vec{k}\vec{r}} = \vec{H}_0 e^{ik\vec{n}\vec{r}} = +5ie^{i2(y+\sqrt{3}z)}\vec{x}$$

$$\vec{H}_0 = +5i\vec{x}$$

$$k\vec{n}\vec{r} = 2\left(y + \sqrt{3}z\right)$$

$\vec{n} = \alpha\vec{x} + \beta\vec{y} + \gamma\vec{z}$ and $\vec{r} = x\vec{x} + y\vec{y} + z\vec{z}$ $\Rightarrow$ $\vec{n}\vec{r} = (\alpha\vec{x} + \beta\vec{y} + \gamma\vec{z}) \cdot (x\vec{x} + y\vec{y} + z\vec{z}) = \alpha x + \beta y + \gamma z$

$k\vec{n}\vec{r} = 2y + 2\sqrt{3}z$ $\Rightarrow$ $k(\alpha x + \beta y + \gamma z) = 2y + 2\sqrt{3}z$

$$k\alpha = 0 \quad \Rightarrow \quad \alpha = 0$$

$$\left.\begin{aligned} k\beta &= 2 \\ k\gamma &= 2\sqrt{3} \end{aligned}\right\} \quad \Rightarrow \quad k^2(\beta^2 + \gamma^2) = 4 + 12 = 16$$

$$|\vec{n}| = \sqrt{\overset{0}{\overbrace{\alpha^2}} + \beta^2 + \gamma^2} = 1 \quad \Rightarrow \quad \beta^2 + \gamma^2 = 1$$

$$k^2(\beta^2 + \gamma^2) = 16 \quad \Rightarrow \quad k^2 = \frac{16}{(\beta^2 + \gamma^2)} = \frac{16}{1} = 16 \quad \Rightarrow \quad \boxed{k = \sqrt{16} = 4 \text{ rad/s}}$$

$$k\beta = 2 \quad \Rightarrow \quad \boxed{\beta = \frac{2}{k} = \frac{2}{4} = 0.5}$$

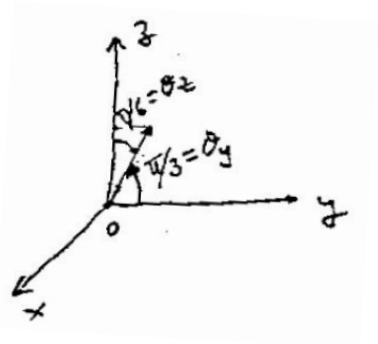
$$k\gamma = 2\sqrt{3} \quad \Rightarrow \quad \boxed{\gamma = \frac{2\sqrt{3}}{k} = \frac{2\sqrt{3}}{4} = 0.5\sqrt{3}}$$

$$\vec{n} = \overset{0}{\overbrace{\alpha}}\vec{x} + \beta\vec{y} + \gamma\vec{z} = 0.5\vec{y} + 0.5\sqrt{3}\vec{z}$$

$$\alpha = 0 = cos\theta_x \quad \Rightarrow \quad \theta_x = \pi/2$$

$$\beta = 0.5 = cos\theta_y \quad \Rightarrow \quad \theta_y = \pi/3$$

$$\gamma = 0.5\sqrt{3} = cos\theta_z \quad \Rightarrow \quad \theta_z = \pi/6$$



c)

$$k^2 = \omega^2\varepsilon\mu + i\sigma\omega\mu = 16 \quad \Rightarrow \quad i\sigma\omega\mu = 0 \quad \Rightarrow \quad \sigma = 0$$

$$\omega^2\varepsilon_r\varepsilon_0\mu_0 = 16 \quad \Rightarrow \quad \varepsilon_r = \frac{16}{\omega^2\varepsilon_0\mu_0} = 16\frac{c_0^2}{\omega^2} = 16\frac{(3.10^8)^2}{(3.10^8)^2} \quad \Rightarrow \quad \boxed{\varepsilon_r = 16}$$

$$Z = \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_0}{\varepsilon_r\varepsilon_0}} = \sqrt{\frac{1}{\varepsilon_r}}\sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{\frac{1}{\varepsilon_r}}Z_0 = \sqrt{\frac{1}{16}}120\pi = \frac{120\pi}{4} \quad \Rightarrow \quad \boxed{Z = 30\pi\ \Omega}$$

d)

$$\vec{H} = \frac{1}{Z}(\vec{n} \times \vec{E}) \quad \Rightarrow \quad \vec{E} = -Z(\vec{n} \times \vec{H})$$

$$\vec{E}(y,z) = -Z\vec{n} \times \vec{H}(y,z) = -30\pi(0.5\vec{y} + 0.5\sqrt{3}\vec{z}) \times \left(+5ie^{i2(y+\sqrt{3}z)}\vec{x}\right) = -i75\pi(-\vec{z} + \sqrt{3}\vec{y})e^{i2(y+\sqrt{3}z)}$$

$$\Rightarrow \quad \boldsymbol{E}(y,z) = i75\pi(-\sqrt{3}\vec{y} + \vec{z})e^{i2(y+\sqrt{3}z)}$$

e)

$$\vec{E} \cdot \vec{H} = 0 \quad \Rightarrow \quad \vec{E} \perp \vec{H}$$

$$\vec{E} \cdot \vec{H} = 0 \quad \Rightarrow \quad \underbrace{\left(i75\pi(-\sqrt{3}\vec{y} + \vec{z})e^{i2(y+\sqrt{3}z)}\right)}_{\vec{E}} \cdot \underbrace{\left(+5ie^{i2(y+\sqrt{3}z)}\vec{x}\right)}_{\vec{H}} \quad \Rightarrow \quad \vec{E} \perp \vec{H}$$

$$\vec{E} \cdot \vec{n} = 0 \quad \Rightarrow \quad \underbrace{\left(i75\pi(-\sqrt{3}\vec{y} + \vec{z})e^{i2(y+\sqrt{3}z)}\right)}_{\vec{E}} \cdot \underbrace{(0.5\vec{y} + 0.5\sqrt{3}\vec{z})}_{\vec{n}} = i75\pi e^{i2(y+\sqrt{3}z)} \underbrace{(0.5\vec{y} + 0.5\sqrt{3}\vec{z})(-\sqrt{3}\vec{y} + \vec{z})}_{\substack{-0.5\sqrt{3}+0.5\sqrt{3} \\ 0}} = 0$$

$$\Rightarrow \quad \vec{E} \perp \vec{n}$$

$$\vec{H} \cdot \vec{n} = 0 \quad \Rightarrow \quad \underbrace{\left(+5ie^{i2(y+\sqrt{3}z)}\vec{x}\right)}_{\vec{H}} \cdot \underbrace{(0.5\vec{y} + 0.5\sqrt{3}\vec{z})}_{\vec{n}} = 0 \quad \Rightarrow \quad \vec{H} \perp \vec{n}$$

An electric field of a monochromatic wave propagating in a free-space given by

$$\vec{E}(x,y,t) = E_0 cos(2\sqrt{2}x - 2\sqrt{2}y - \omega t)\vec{z}$$

where E_0 and ω are given positive constants.

a) Write the complex expression $\vec{E}(x,y)$ of the electric field above.
b) Determine the propagation direction $\vec{n}$ and the frequency (f) of the plane wave.
c) Considering the relation between $\vec{E}$, $\vec{H}$ and $\vec{n}$, find the complex magnetic field $\vec{H}(x,y)$.
d) Prove that this wave is a plane wave or not (Use the relations for $\vec{E}$, $\vec{H}$ and $\vec{n}$)

Solution

a)

$$\vec{E}(x,y,t) = E_0 cos(2\sqrt{2}x - 2\sqrt{2}y - \omega t)\vec{z} = \text{Re}\left\{E_0 e^{i(2\sqrt{2}x - 2\sqrt{2}y - \omega t)}\vec{z}\right\} = \text{Re}\left\{\underbrace{E_0 e^{i(2\sqrt{2}x - 2\sqrt{2}y)}\vec{z}}_{\vec{E}(x,y)} e^{-i\omega t}\right\}$$

$$\boxed{\vec{E}(x,y) = E_0 e^{i2\sqrt{2}(x-y)}\vec{z}}$$

b)

propagation direction

$$\vec{n} = \alpha\vec{x} + \beta\vec{y} + \gamma\vec{z} \;;\; \vec{r} = x\vec{x} + y\vec{y} + z\vec{z}$$

$$\vec{E}(x,y) = \vec{E}_0 e^{ik\vec{n}\vec{r}} = \vec{E}_0 e^{ik(\alpha x + \beta y + \gamma z)} = E_0 e^{i2\sqrt{2}(x-y)}\vec{z}$$

$$\vec{E}_0 = E_0\vec{z}$$

$$ik(\alpha x + \beta y + \gamma z) = i2\sqrt{2}(x-y)$$

$$\left.\begin{aligned} k\alpha &= 2\sqrt{2} \\ k\beta &= -2\sqrt{2} \\ k\gamma &= 0 \;\Rightarrow\; \gamma = 0 \end{aligned}\right\} \Rightarrow \quad k^2\alpha^2 + k^2\beta^2 = 8 + 8 = 16$$

The vector $\vec{n}$ is a unit vector $|\vec{n}| = 1 \;\Rightarrow\; |\vec{n}|^2 = \sqrt{\alpha^2 + \beta^2 + \gamma^2} = 1 \;\Rightarrow\; \alpha^2 + \beta^2 = 1$

$$k^2\alpha^2 + k^2\beta^2 = k^2\underbrace{(\alpha^2 + \beta^2)}_{1} = 16 \;\Rightarrow\; k^2 = 16 \;\Rightarrow\; k = 4 \text{ rad/m}$$

$$\left.\begin{aligned} k\alpha = 2\sqrt{2} \;&\Rightarrow\; \alpha = \frac{2\sqrt{2}}{4} = \frac{\sqrt{2}}{2} \\ k\beta = -2\sqrt{2} \;&\Rightarrow\; \beta = -\frac{2\sqrt{2}}{4} = -\frac{\sqrt{2}}{2} \\ k\gamma = 0 \;&\Rightarrow\; \gamma = 0 \end{aligned}\right\} \Rightarrow \quad \boxed{\vec{n} = \frac{\sqrt{2}}{2}\vec{x} - \frac{\sqrt{2}}{2}\vec{y}}$$

Calculation of the frequency: $k = \omega/c = \omega\sqrt{\varepsilon_0\mu_0} = 4 \;\Rightarrow\; f = \frac{\omega}{2\pi} = \frac{4}{2\pi\sqrt{\varepsilon_0\mu_0}} = \frac{4c}{2\pi} = \frac{2}{\pi}3.10^8 = \boxed{\frac{600}{\pi} \cong 191 \text{ MHz}}$

c)

$$\vec{H} = \frac{1}{Z}(\vec{n} \times \vec{E}) = \frac{1}{120\pi}\left(\frac{\sqrt{2}}{2}\vec{x} - \frac{\sqrt{2}}{2}\vec{y}\right) \times E_0 e^{i2\sqrt{2}(x-y)}\vec{z} = \boxed{-\frac{\sqrt{2}E_0}{240\pi}(\vec{y} + \vec{x})e^{i2\sqrt{2}(x-y)}}$$

d)

$$\vec{H}\cdot\vec{n} = \left[-\frac{\sqrt{2}}{240\pi}(\vec{y} + \vec{x})E_0 e^{i2\sqrt{2}(x-y)}\right]\cdot\left[\frac{\sqrt{2}}{2}\vec{x} - \frac{\sqrt{2}}{2}\vec{y}\right] = -\frac{E_0}{240\pi}e^{i2\sqrt{2}(x-y)}(1-1) = 0$$

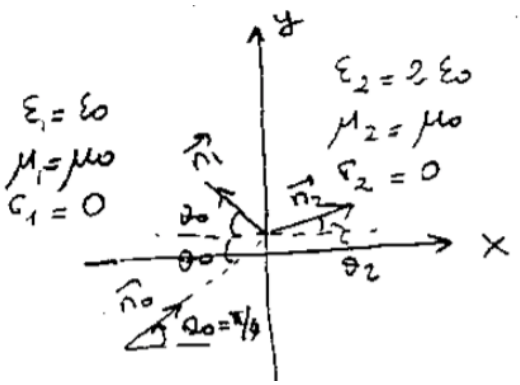
$$\left.\begin{aligned} \vec{H}\cdot\vec{n} = 0 \;&\Rightarrow\; \vec{H} \perp \vec{n} \\ \vec{E}\cdot\vec{n} = 0 \;&\Rightarrow\; \vec{E} \perp \vec{n} \\ \vec{E}\cdot\vec{H} = 0 \;&\Rightarrow\; \vec{E} \perp \vec{H} \end{aligned}\right\} \Rightarrow \quad \boxed{\vec{E} \perp \vec{H} \perp \vec{n}}$$

Let the permittivities of the lossless and nonmagnetic materials filling the half-spaces $x < 0$ and $x > 0$ be $\varepsilon_1 = \varepsilon_0$ and $\varepsilon_2 = 2\varepsilon_0$, respectively. Assume that a monochromatic plane wave excited in the region $x < 0$ has the magnetic field given as follows

$$H_x = 0 \;\; ; \;\; H_y = 0 \;\; ; \;\; H_z = cos\left(\frac{x+y}{\sqrt{2}} - 3.10^8 t\right).$$

a) Find the frequency, phase velocity and wavelength of this (incident) wave.
b) Determine the propagation directions and equiphase surfaces associated with the incident and reflected waves.
c) Find the propagation direction and amplitude of the transmitted wave.
d) Compute the average power transported by the reflected and refracted waves through an area S lying on the plane $x = 0$ and compare them with what is transported by the incident wave through the same area.

Solution



a) Let us calculate the complex notation of $\vec{H}_0$

$$\vec{H}_0(x,y,t) = H_0\vec{z} = cos\left(\frac{x+y}{\sqrt{2}} - 3.10^8 t\right)\vec{z} = \text{Re}\left\{H_0 e^{i\left(\frac{x+y}{\sqrt{2}} - 3.10^8 t\right)}\vec{z}\right\} = \text{Re}\left\{\underbrace{H_0 e^{i\left(\frac{x+y}{\sqrt{2}}\right)}}_{H_0(x,y)} e^{-i3.10^8 t}\vec{z}\right\}$$

$$\boxed{\vec{H}_0(x,y) = H_0 e^{i\left(\frac{x+y}{\sqrt{2}}\right)}\vec{z}}$$

$$e^{-i3.10^8 t} = e^{-i\omega t} \;\; \Rightarrow \;\; \omega = 3.10^8 \text{ rad/s} \;\; \Rightarrow \;\; \boxed{f = \frac{\omega}{2\pi} = \frac{300}{2\pi} \text{ MHz}}$$

$$v_{phase} = \frac{\omega}{|\nabla\alpha|} = \frac{\omega}{\left|\nabla\left(\frac{x+y}{\sqrt{2}}\right)\right|} = \frac{\omega}{\left|\frac{1}{\sqrt{2}}\vec{x} + \frac{1}{\sqrt{2}}\vec{y}\right|} = \frac{\omega}{|1|} = \omega = 3.10^8 \text{ rad/s}$$

$$\lambda = v_{phase}T = \frac{v_{phase}}{f} = \frac{\omega}{f} = \frac{3.10^8 \text{ rad/s}}{\frac{3.10^8 \text{ rad/s}}{2\pi}} = 2\pi \text{ m}$$

b) The propagation direction of the incident wave

$k = \omega\sqrt{\varepsilon\mu}$

$$k_{incident} = k_1 = \omega\sqrt{\varepsilon_1\mu_1} = \omega\sqrt{\varepsilon_0\mu_0} = \frac{\omega}{c_0} = \frac{3.10^8 \text{ rad/s}}{3.10^8 \text{ m/s}} = 1 \text{ rad/m}$$

$$e^{ik_1\vec{n}_0\vec{r}} = e^{i\overbrace{k_1}^{1}(cos\theta_0\vec{x} + sin\theta_0\vec{y})(x\vec{x}+y\vec{y})} = e^{i(xcos\theta_0 + ysin\theta_0)} = e^{i\left(\frac{x+y}{\sqrt{2}}\right)}$$

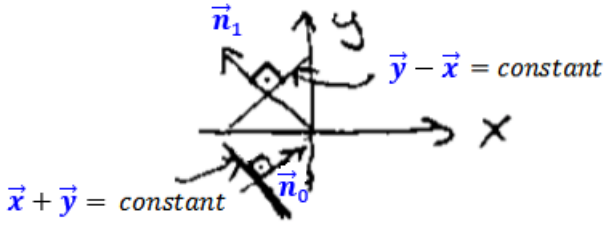
$$xcos\theta_0 = \frac{x}{\sqrt{2}} \;\; \Rightarrow \;\; cos\theta_0 = \frac{1}{\sqrt{2}} \;\; \Rightarrow \;\; \theta_0 = \frac{\pi}{4}$$

$$ysin\theta_0 = \frac{y}{\sqrt{2}} \;\; \Rightarrow \;\; sin\theta_0 = \frac{1}{\sqrt{2}} \;\; \Rightarrow \;\; \theta_0 = \frac{\pi}{4}$$

$$\vec{n}_0 = cos\theta_0\vec{x} + sin\theta_0\vec{y} = cos\left(\frac{\pi}{4}\right)\vec{x} + sin\left(\frac{\pi}{4}\right)\vec{y} \quad \Rightarrow \quad \vec{n}_0 = \frac{1}{\sqrt{2}}\vec{x} + \frac{1}{\sqrt{2}}\vec{y}$$

The equiphase surface of the incident wave

$$\vec{n}_0 = \frac{1}{\sqrt{2}}\vec{x} + \frac{1}{\sqrt{2}}\vec{y} = constant \quad \Rightarrow \quad \vec{x} + \vec{y} = constant$$



The propagation direction of the reflected wave

$$\vec{n}_1 = -\frac{1}{\sqrt{2}}\vec{x} + \frac{1}{\sqrt{2}}\vec{y}$$

The equiphase surface of the reflected field

$$\vec{n}_1 = -\frac{1}{\sqrt{2}}\vec{x} + \frac{1}{\sqrt{2}}\vec{y} = constant \quad \Rightarrow \quad \vec{y} - \vec{x} = constant$$

c) The propagation direction of the transmitted wave

$$\vec{H}_2(x,y) = H_2 e^{ik_2\vec{n}_2\vec{r}}\vec{z} = H_2 e^{ik_2(xcos\theta_2 + ysin\theta_2)}\vec{z}$$

Using Snell law

$$k_1 sin\theta_0 = k_2 sin\theta_2 \quad \Rightarrow \quad sin\theta_2 = \frac{k_1}{k_2}sin\theta_0 = \frac{\omega\sqrt{\varepsilon_0\mu_0}}{\omega\sqrt{\varepsilon_2}\sqrt{\varepsilon_0\mu_0}}sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}} = \frac{1}{2} \quad \Rightarrow \quad \theta_2 = \frac{\pi}{6}$$

$$\vec{n}_2 = cos\left(\frac{\pi}{6}\right)\vec{x} + sin\left(\frac{\pi}{6}\right)\vec{y} = \frac{\sqrt{3}}{2}\vec{x} + \frac{1}{2}\vec{y}$$

Due to the amplitude of the incident wave is one Transmission coefficient

$$H_2 = \mathrm{T} = \frac{2Z_1 cos\theta_0}{Z_1 cos\theta_0 + Z_2 cos\theta_2} = \frac{2Z_1 cos\theta_0}{Z_1 cos\theta_0\left(1 + \frac{Z_2}{Z_1}\frac{cos\theta_2}{cos\theta_0}\right)} = \frac{2}{1 + \frac{Z_2}{Z_1}\frac{cos\theta_2}{cos\theta_0}} = \frac{2}{1 + \frac{\sqrt{\frac{\mu_0}{2\varepsilon_0}}cos\left(\frac{\pi}{6}\right)}{\sqrt{\frac{\mu_0}{\varepsilon_0}}cos\left(\frac{\pi}{4}\right)}}$$

$$H_2 = \mathrm{T} = \frac{2}{1 + \frac{\sqrt{\frac{\mu_0}{2\varepsilon_0}}cos\left(\frac{\pi}{6}\right)}{\sqrt{\frac{\mu_0}{\varepsilon_0}}cos\left(\frac{\pi}{4}\right)}} = \frac{2}{1 + \frac{1}{\sqrt{2}}\frac{\frac{\sqrt{3}}{2}}{\frac{1}{\sqrt{2}}}} = \frac{2}{1 + \frac{\sqrt{3}}{2}} = \frac{4}{2 + \sqrt{3}}$$

d) For the transmitted wave

$$\vec{P}_2 = \mathrm{Re}\{\vec{P}_{complex2}\} = \frac{1}{2}\mathrm{Re}\{\vec{E}_2 \times \vec{H}_2^*\} = \frac{1}{2}\mathrm{Re}\{-Z_2(\vec{n}_2 \times \vec{H}_2) \times \vec{H}_2^*\}$$

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{A} \cdot \vec{C}) - \vec{C} \cdot (\vec{A} \cdot \vec{B})$$

$$(\vec{n}_2 \times \vec{H}_2) \times \vec{H}_2^* = -\vec{H}_2^* \times (\vec{n}_2 \times \vec{H}_2) = -\left[\vec{n}_2 \cdot (\vec{H}_2^* \cdot \vec{H}_2) - \vec{H}_2 \cdot \overbrace{(\vec{H}_2^* \cdot \vec{n}_2)}^{0}\right] = -|\vec{H}_2|^2\vec{n}_2$$

$$\Rightarrow \quad \vec{P}_2 = \mathrm{Re}\{\vec{P}_{complex2}\} = \frac{1}{2}\mathrm{Re}\{\vec{E}_2 \times \vec{H}_2^*\} = \frac{1}{2}\mathrm{Re}\{-Z_2(\vec{n}_2 \times \vec{H}_2) \times \vec{H}_2^*\} = \frac{1}{2}Z_2|\vec{H}_2|^2\vec{n}_2$$

$$\vec{P}_2 = \frac{1}{2} Z_2 |\vec{H}_2|^2 \vec{n}_2 = \frac{1}{2}\sqrt{\frac{\mu_0}{2\varepsilon_0}} \left| H_{02} e^{ik_2(x\cos\theta_2 + y\sin\theta_2)} \right|^2 \left(\frac{\sqrt{3}}{2}\vec{x} + \frac{1}{2}\vec{y}\right)$$

$$\vec{P}_2 = \frac{1}{2}\frac{1}{\sqrt{2}} 120\pi \left| \left(\frac{4}{2+\sqrt{3}}\right) e^{ik_2(x\cos\theta_2 + y\sin\theta_2)} \right|^2 \left(\frac{\sqrt{3}}{2}\vec{x} + \frac{1}{2}\vec{y}\right)$$

$$\vec{P}_2 = \frac{1}{2}\frac{1}{\sqrt{2}} 120\pi \left(\frac{4}{2+\sqrt{3}}\right)^2 \overbrace{\left| e^{ik_2(x\cos\theta_2 + y\sin\theta_2)} \right|^2}^{1} \left(\frac{\sqrt{3}}{2}\vec{x} + \frac{1}{2}\vec{y}\right) = \frac{1}{2}\frac{1}{\sqrt{2}} 120\pi \frac{16}{(2+\sqrt{3})^2} \left(\frac{\sqrt{3}}{2}\vec{x} + \frac{1}{2}\vec{y}\right)$$

$$W_2 = \int_S \vec{P}_2 \cdot \vec{ds} = \int_S \frac{1}{2}\frac{1}{\sqrt{2}} 120\pi \frac{16}{(2+\sqrt{3})^2} \left(\frac{\sqrt{3}}{2}\vec{x} + \frac{1}{2}\vec{y}\right) \cdot ds\vec{x} = \frac{1}{2}\frac{1}{\sqrt{2}} 120\pi \frac{16}{(2+\sqrt{3})^2} \frac{\sqrt{3}}{2} \underbrace{\int_S ds}_{S} = \frac{240\sqrt{6}}{(2+\sqrt{3})^2}\pi S$$

For the reflected wave

$$\vec{P}_1 = \mathrm{Re}\{\vec{P}_{complex1}\} = \frac{1}{2}\mathrm{Re}\{\vec{E}_1 \times \vec{H}_1^*\} = \frac{1}{2}\mathrm{Re}\{-Z_1(\vec{n}_1 \times \vec{H}_1) \times \vec{H}_1^*\} = \frac{1}{2} Z_1 |\vec{H}_1|^2 \vec{n}_1$$

$\vec{E}_1 = -Z_1(\vec{n}_1 \times \vec{H}_1)$

$$1 + \overbrace{|\vec{H}_1|}^{\Gamma} = \overbrace{|\vec{H}_2|}^{T} \quad \Rightarrow \quad |\vec{H}_1| = 1 - |\vec{H}_2| = 1 - \frac{4}{2+\sqrt{3}} = \frac{2-\sqrt{3}}{2+\sqrt{3}}$$

$$\vec{P}_1 = \frac{1}{2} Z_1 |\vec{H}_1|^2 \vec{n}_1 = \frac{1}{2}\sqrt{\frac{\mu_0}{\varepsilon_0}} \left(\frac{2-\sqrt{3}}{2+\sqrt{3}}\right)^2 \left(-\frac{1}{\sqrt{2}}\vec{x} + \frac{1}{\sqrt{2}}\vec{y}\right) = \frac{120\pi}{2} \left(\frac{2-\sqrt{3}}{2+\sqrt{3}}\right)^2 \left(-\frac{1}{\sqrt{2}}\vec{x} + \frac{1}{\sqrt{2}}\vec{y}\right)$$

$$W_1 = \int_S \vec{P}_1 \cdot (-\vec{ds}) = \int_S \frac{120\pi}{2} \left(\frac{2-\sqrt{3}}{2+\sqrt{3}}\right)^2 \left(-\frac{1}{\sqrt{2}}\vec{x} + \frac{1}{\sqrt{2}}\vec{y}\right) \cdot ds(-\vec{x}) = \frac{60\pi}{\sqrt{2}} \left(\frac{2-\sqrt{3}}{2+\sqrt{3}}\right)^2 \underbrace{\int_S ds}_{S} = \frac{60\pi}{\sqrt{2}} \left(\frac{2-\sqrt{3}}{2+\sqrt{3}}\right)^2 S$$

For the incident wave

$$\vec{P}_0 = \mathrm{Re}\{\vec{P}_{complex0}\} = \frac{1}{2}\mathrm{Re}\{\vec{E}_0 \times \vec{H}_0^*\} = \frac{1}{2}\mathrm{Re}\{-Z_0(\vec{n}_0 \times \vec{H}_0) \times \vec{H}_0^*\} = \frac{1}{2} Z_0 \overbrace{|\vec{H}_0|^2}^{1} \vec{n}_0$$

$$\vec{P}_0 = \frac{1}{2} Z_0 \vec{n}_0 = \frac{1}{2}\sqrt{\frac{\mu_0}{\varepsilon_0}} \left(\frac{1}{\sqrt{2}}\vec{x} + \frac{1}{\sqrt{2}}\vec{y}\right) = \frac{120\pi}{2} \left(\frac{1}{\sqrt{2}}\vec{x} + \frac{1}{\sqrt{2}}\vec{y}\right) = 60\pi \left(\frac{1}{\sqrt{2}}\vec{x} + \frac{1}{\sqrt{2}}\vec{y}\right)$$

$$W_0 = \int_S \vec{P}_0 \cdot \vec{ds} = \int_S 60\pi \left(\frac{1}{\sqrt{2}}\vec{x} + \frac{1}{\sqrt{2}}\vec{y}\right) \cdot ds\vec{x} = \frac{60\pi}{\sqrt{2}} \underbrace{\int_S ds}_{S} = \frac{60\pi}{\sqrt{2}} S$$

For all the waves

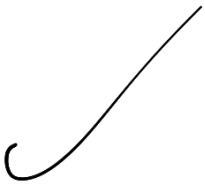
$$W_0 = W_1 + W_2$$

$$\frac{60\pi}{\sqrt{2}} S = \frac{60\pi}{\sqrt{2}} \left(\frac{2-\sqrt{3}}{2+\sqrt{3}}\right)^2 S + \frac{240\sqrt{6}}{(2+\sqrt{3})^2}\pi S = \frac{60\pi}{\sqrt{2}} \left[\frac{(2-\sqrt{3})^2}{(2+\sqrt{3})^2} + \frac{\sqrt{2}\times 4\sqrt{6}}{(2+\sqrt{3})^2}\right] S$$

$$\frac{60\pi}{\sqrt{2}}S=\frac{60\pi}{\sqrt{2}}\left[\frac{(2-\sqrt{3})^2+\overbrace{\sqrt{2}\times 4\sqrt{6}}^{8\sqrt{3}}}{(2+\sqrt{3})^2}\right]S=\frac{60\pi}{\sqrt{2}}\left[\frac{7-4\sqrt{3}+8\sqrt{3}}{(2+\sqrt{3})^2}\right]S=\frac{60\pi}{\sqrt{2}}\left[\frac{7+4\sqrt{3}}{(2+\sqrt{3})^2}\right]S$$

$$\frac{60\pi}{\sqrt{2}}S=\frac{60\pi}{\sqrt{2}}\left[\frac{(2-\sqrt{3})^2+\overbrace{\sqrt{2}\times 4\sqrt{6}}^{8\sqrt{3}}}{(2+\sqrt{3})^2}\right]S=\frac{60\pi}{\sqrt{2}}\left[\frac{7-4\sqrt{3}+8\sqrt{3}}{(2+\sqrt{3})^2}\right]S=\frac{60\pi}{\sqrt{2}}\left[\frac{\overbrace{(2+\sqrt{3})^2}^{1}}{(2+\sqrt{3})^2}\right]S$$

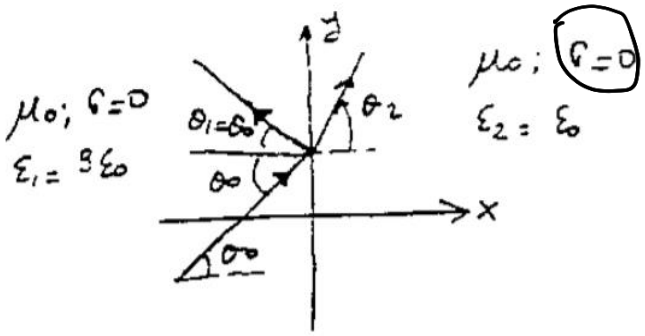
$$\frac{60\pi}{\sqrt{2}}S=\frac{60\pi}{\sqrt{2}}S \quad \Rightarrow \quad \boxed{W_0=W_1+W_2}$$



Suppose that the regions $x<0$ and $x>0$ are filled with nonconducting and nonmagnetic simple dielectrics of parameters $\varepsilon_1=9\varepsilon_0$ and $\varepsilon_2=\varepsilon_0$, respectively while a plane wave of frequency $f=300/2\pi$ MHz propagating in the direction $\vec{n}_0(\sqrt{3}/2,1/2,0)$ is excited in the region.

a) Find the equiphase surfaces as well as the phase velocity of the wave propagating into the region $x>0$.
b) Let the amplitude of any field component of the refracted wave at the plane $x=0$ be A_2. What will be the values of the same amplitude at the planes $x=1$ and $x=2$?
c) Compute the densities of power stored in the region $x>0$, on the planes $x=2$ and $x=3$ in terms of decibel.

Solution



a)

$$\vec{n}_0=\frac{\sqrt{3}}{2}\vec{x}+\frac{1}{2}\vec{y}=cos\theta_0\vec{x}+sin\theta_0\vec{y} \quad\Rightarrow\quad \left.\begin{matrix} cos\theta_0=\frac{\sqrt{3}}{2} \\ sin\theta_0=\frac{1}{2}\end{matrix}\right\} \quad\Rightarrow\quad \theta_0=\frac{\pi}{6}$$

Using Snell law

$$k_1sin\theta_0=k_2sin\theta_2 \quad\Rightarrow\quad sin\theta_2=\frac{k_1}{k_2}sin\theta_0=\frac{\omega\sqrt{\varepsilon_1\mu_0}}{\omega\sqrt{\varepsilon_2\mu_0}}\overbrace{sin\left(\frac{\pi}{6}\right)}^{1/2}=\frac{3\sqrt{\varepsilon_0}}{\sqrt{\varepsilon_0}}\times\frac{1}{2}=\frac{3}{2}>1 \quad\Rightarrow\quad \theta_2=complex\ angle\ !!!$$

$$U_2\approx e^{ik_2\vec{n}_2\vec{r}}=e^{ik_2\overbrace{(cos\theta_2\vec{x}+sin\theta_2\vec{y})}^{\vec{n}_2}\overbrace{(x\vec{x}+y\vec{y})}^{\vec{r}}}=e^{ik_2(xcos\theta_2+ysin\theta_2)}=e^{ik_2\left(x\sqrt{1-\sin^2\theta_2}+ysin\theta_2\right)}$$

$$U_2\approx e^{-\beta x}e^{ik_2\tilde{n}_2\vec{r}}=e^{ik_2\left(x\sqrt{1-\frac{9}{4}}+y\frac{3}{2}\right)}=e^{i\frac{k_2}{2}(ix\sqrt{5}+y3)}=e^{-k_2\frac{\sqrt{5}}{2}x}e^{ik_2\frac{3}{2}y}$$

$$\tilde{n}_2=\vec{y} \quad\Rightarrow\quad k_2\frac{3}{2}y=constant \quad\Rightarrow\quad \boxed{y=constant}$$: the equiphase surfaces

$$\alpha(y)=k_2\frac{3}{2}y \quad\Rightarrow\quad |\nabla\alpha(y)|=\frac{d\alpha}{dy}\vec{y}=k_2\frac{3}{2}\vec{y}$$

$$v_{phase}=\frac{\omega}{|\nabla\alpha|}=\frac{\omega}{k_2\frac{3}{2}}=\frac{\omega}{\omega\sqrt{\varepsilon_0\mu_0}\frac{3}{2}}=\frac{2}{3\sqrt{\varepsilon_0\mu_0}}=\frac{2}{3}c_0=2.10^8\ \text{m/s}$$

b)

$$k_2=\omega\sqrt{\varepsilon_2\mu_2}=\omega\sqrt{\varepsilon_0\mu_0}=\frac{\omega}{c_0}=\frac{2\pi f}{c_0}=\frac{2\pi\frac{300}{2\pi}.10^6\ \text{rad/s}}{3.10^8\ \text{m/s}}=1\ \text{rad/m}$$

$$U_2=Ae^{-k_2\frac{\sqrt{5}}{2}x}e^{ik_2\frac{3}{2}y} \quad\Rightarrow\quad |U_2|_{x=0}|=A_2$$

$$U_2|_{x=1}=Ae^{-k_2\frac{\sqrt{5}}{2}1}e^{ik_2\frac{3}{2}y} \quad\Rightarrow\quad |U_2|_{x=1}|=A_2e^{-\frac{\sqrt{5}}{2}}$$

$$U_2|_{x=2}=Ae^{-k_2\frac{\sqrt{5}}{2}2}e^{ik_2\frac{3}{2}y} \quad\Rightarrow\quad |U_2|_{x=2}|=A_2e^{-\sqrt{5}}$$

The wave amplitude is decreasing along $+x$ propagation.

c)

$$W_2 = |U_2|_{x=2}|^2 = A^2\left(e^{-\frac{\sqrt{5}}{2}2}\right)^2 = A_2^2 e^{-2\sqrt{5}}$$

$$W_3 = |U_2|_{x=3}|^2 = A^2\left(e^{-\frac{\sqrt{5}}{2}3}\right)^2 = A_2^2 e^{-3\sqrt{5}}$$

$$\Rightarrow \quad Comparison = 10\log_{10}\left(\frac{W_2}{W_3}\right) = 10\log_{10}\left(\frac{A_2^2 e^{-2\sqrt{5}}}{A_2^2 e^{-3\sqrt{5}}}\right) = 10\log_{10}\left(e^{+\sqrt{5}}\right) = 10\sqrt{5}\log_{10}(e) = 9.71 \text{ dB}$$

Vizebarajaya kadar

4.2.6 – Refraction to Lossy Medium

Let the electromagnetic constitutive parameters of the media $x < 0$ and $x > 0$ be $\varepsilon_1 = 12\varepsilon_0, \mu_1 = 0, \sigma_1 = 0$ and $\varepsilon_2 = 2\varepsilon_0, \mu_2 = 0, \sigma_2 \neq 0$, respectively. A monochromatic plane wave has the frequency $f = 150/\pi$ MHz and propagates in the direction $\vec{n}_0(\sqrt{3}/2,1/2,0)$ in the region $x < 0$ causes a refracted wave propagating along the direction $\vec{n}_2(1/2,\sqrt{3}/2,0)$ in the region $x > 0$. Determine the conductivity σ_2 of the medium $x > 0$.

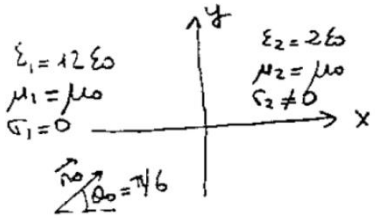
$x < 0$ ve $x > 0$ ile ayrılan iki bölgenin elektromanyetik bünye bağıntıları $\varepsilon_1 = 12\varepsilon_0, \mu_1 = 0, \sigma_1 = 0$ ve $\varepsilon_2 = 2\varepsilon_0, \mu_2 = 0, \sigma_2 \neq 0$ olarak verilmiştir. $f = 150/\pi$ MHz frekansına sahip ve $x < 0$ bölgesinde $\vec{n}_0(\sqrt{3}/2,1/2,0)$ yönünde yayılan monokromatik bir düzlem dalga $x > 0$ bölgesinde $\vec{n}_2(1/2,\sqrt{3}/2,0)$ yönünde kırılan (refracted) bir dalgaya sebep olmaktadır. Buna göre $x > 0$ bölgesinin σ_2 iletkenlik değerini hesaplayınız.

Solution

a)

$\vec{n}_2 = \cos\theta_2 \vec{x} + \sin\theta_2 \vec{y}$

$$\vec{n}_0 = cos\theta_0\vec{x} + sin\theta_0\vec{y} = \frac{\sqrt{3}}{2}\vec{x} + \frac{1}{2}\vec{y} \quad \Rightarrow \quad cos\theta_0 = \frac{\sqrt{3}}{2} \;\; ; \;\; sin\theta_0 = \frac{1}{2} \quad \Rightarrow \quad \boxed{\theta_0 = \frac{\pi}{6}}$$



Using Snell law

$$k_1 sin\theta_0 = k_2 sin\theta_2 \quad \Rightarrow \quad k_2 sin\theta_2 = \overbrace{2\pi f\sqrt{12\varepsilon_0\mu_0}}^{\omega\sqrt{\varepsilon_1\mu_1}} sin\theta_0 = 2\pi\frac{150}{\pi}10^6 2\sqrt{3}\frac{1}{3.10^8}\frac{1}{2} = \sqrt{3} \quad \Rightarrow \quad k_2^2 \sin^2\theta_2 = 3$$

$\cos\theta_2 = \sqrt{1-\sin^2\theta_2}$

$$\Rightarrow \quad k_2 cos\theta_2 = \sqrt{k_2^2 - k_2^2\sin^2\theta_2} = \sqrt{(\omega^2\varepsilon_2\mu_2 + i\omega\sigma_2\mu_2) - 3} = \alpha + i\beta \quad ; \quad \alpha, \beta > 0$$

bir kaç matematik bilmiyorsa

birim vektör $\frac{\vec{n}}{|\vec{n}|}$

$$u_2(x,y) = A_2 e^{ik_2(cos\theta_2 x + sin\theta_2 y)} = A_2 e^{i(k_2 cos\theta_2 x + k_2 sin\theta_2 y)} = A_2 e^{i(\alpha+i\beta)x} e^{i\sqrt{3}y} = A_2 e^{-\beta x} e^{i(\alpha x + \sqrt{3}y)}$$

normalizasyon

$$u_2(x,y) = A_2 e^{-\beta x} e^{i(\alpha x+\sqrt{3}y)} = A_2 e^{-\beta x} e^{i\sqrt{\alpha^2+3}\left(\frac{\alpha}{\sqrt{\alpha^2+3}}x + \frac{\sqrt{3}}{\sqrt{\alpha^2+3}}y\right)} = A_2 e^{-\beta x} e^{i\sqrt{\alpha^2+3}(cos\hat{\theta}_2 x + sin\hat{\theta}_2 y)}$$

$e^{i|k_2|\vec{n}_2\cdot\vec{r}}$

$\vec{n}_2 = \cos\theta_2\hat{x} + \sin\theta_2\hat{y}$

$\vec{r} = x\hat{x} + y\hat{y}$

$$\left.\begin{aligned} cos\hat{\theta}_2 &= \frac{\alpha}{\sqrt{\alpha^2+3}} \\ sin\hat{\theta}_2 &= \frac{\sqrt{3}}{\sqrt{\alpha^2+3}} \end{aligned}\right\} \Rightarrow \cos^2\hat{\theta}_2 + \sin^2\hat{\theta}_2 = 1$$

$$\vec{n}_2 = cos\hat{\theta}_2\vec{x} + sin\hat{\theta}_2\vec{y} = \frac{1}{2}\vec{x} + \frac{\sqrt{3}}{2}\vec{y} \quad \Rightarrow \quad cos\hat{\theta}_2 = \frac{1}{2} \;\; ; \;\; sin\hat{\theta}_2 = \frac{\sqrt{3}}{2}$$

$$\left.\begin{aligned} cos\hat{\theta}_2 &= \frac{\alpha}{\sqrt{\alpha^2+3}} = \frac{1}{2} \\ sin\hat{\theta}_2 &= \frac{\sqrt{3}}{\sqrt{\alpha^2+3}} = \frac{\sqrt{3}}{2} \end{aligned}\right\} \Rightarrow \boxed{\alpha = 1}$$

$$\left(\sqrt{(\omega^2\varepsilon_2\mu_2 + i\omega\sigma_2\mu_2) - 3}\right)^2 = (\alpha + i\beta)^2 \Rightarrow \quad \omega^2\varepsilon_2\mu_2 + i\omega\sigma_2\mu_2 - 3 = \alpha^2 - \beta^2 + i2\alpha\beta$$

$$\alpha^2 - \beta^2 = \omega^2\varepsilon_2\mu_2 - 3 \quad \Rightarrow \quad \beta^2 = \alpha^2 + 3 - \omega^2\varepsilon_2\mu_2 = 1 + 3 - \left(2\pi\frac{150}{\pi}10^6\right)^2 2\frac{1}{36\pi}10^{-9}4\pi10^{-7} = 2$$

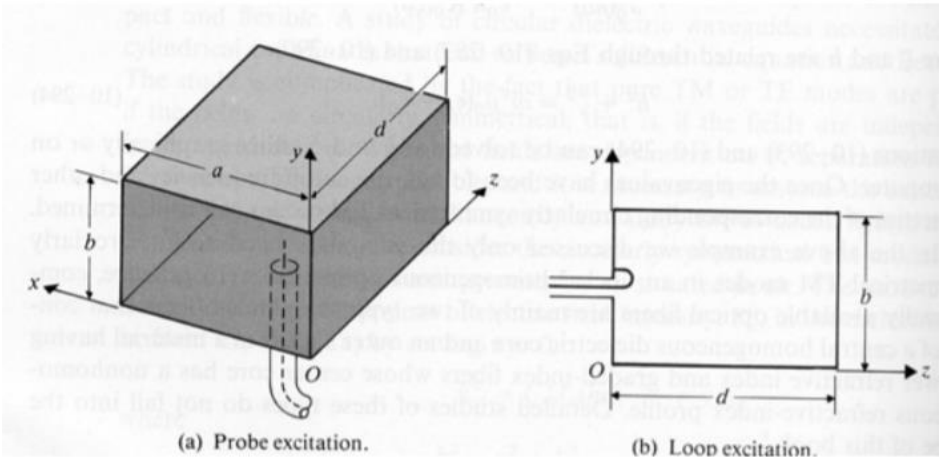
$$\beta^2 = 2 \quad \Rightarrow \quad \beta = \sqrt{2}$$

$$2\alpha\beta = \omega\sigma_2\mu_2 \quad \Rightarrow \quad \sigma_2 = \frac{2\alpha\beta}{\omega\mu_2} = \frac{2\alpha\beta}{2\pi f\mu_0} = \frac{2\times1\times\sqrt{2}}{2\pi\frac{150}{\pi}10^6 4\pi10^{-7}} \quad \Rightarrow \quad \boxed{\sigma_2 = \frac{1}{30\pi\sqrt{2}} \text{ S/m}}$$

A rectangular resonator with dimensions of a $= 4.755$ cm and b $= 2.215$ cm is filled with polyethylene ($\varepsilon_r = 2.25$, $\mu_r = 1$ material (shown below). To occur a resonance at f $= 5$ GHz for a mode number *(m, n, p)* = (1,0,1)

a) Calculate the length (d) of the resonator.

b) Prove that there is a $\pi/2$ phase difference between $\vec{\boldsymbol{E}}$ and $\vec{\boldsymbol{H}}$ fields (as an example, for TE^z_{101} mode).



Solution

a) The resonance frequency of the rectangular resonator is

$$f = \frac{1}{2\pi}\frac{1}{\sqrt{\varepsilon_r\varepsilon_0\mu_r\mu_0}}\sqrt{\left(\frac{m\pi}{a}\right)^2+\left(\frac{n\pi}{b}\right)^2+\left(\frac{p\pi}{d}\right)^2}$$

where if a resonance is at f $= 5$ GHz with the mode number *(m, n, p)* = (1,0,1)

$$5\text{ GHz} = \frac{1}{2\pi}\underbrace{\frac{1}{\sqrt{\varepsilon_0\mu_0}}}_{c_0=3\times10^8\text{ m/s}}\frac{1}{\sqrt{2.25}}\sqrt{\left(\frac{\pi}{4.755}\right)^2+\left(\frac{\pi}{d}\right)^2} \Rightarrow d = 2.20\text{ cm}$$

b) For the general TE^z_{mnp} modes, the resonator fields are

$$E_x = -A'\left(\frac{n\pi}{b}\right)\cos\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)\sin\left(\frac{p\pi}{d}z\right)$$

$$E_y = -A'\left(\frac{m\pi}{a}\right)\sin\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)\sin\left(\frac{p\pi}{d}z\right)$$

$$E_z = 0$$

$$H_x = A'\left(\frac{i}{\omega\mu}\right)\left(\frac{m\pi}{a}\right)\left(\frac{p\pi}{d}\right)\sin\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)\cos\left(\frac{p\pi}{d}z\right)$$

$$H_y = A'\left(\frac{i}{\omega\mu}\right)\left(\frac{m\pi}{a}\right)\left(\frac{p\pi}{d}\right)\cos\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)\cos\left(\frac{p\pi}{d}z\right)$$

$$H_z = -A'\left(\frac{i}{\omega\mu}\right)\left[\left(\frac{m\pi}{a}\right)^2+\left(\frac{n\pi}{b}\right)^2\right]\cos\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)\sin\left(\frac{p\pi}{d}z\right)$$

For the mode number (m, n, p) = (1,0,1), the fields become $m=1 \quad n=0 \quad p=1$

$$E_x = 0$$

$$E_y = -A'\left(\frac{\pi}{a}\right) sin\left(\frac{\pi}{a}x\right) sin\left(\frac{\pi}{d}z\right)$$

$$E_z = 0$$

$$H_x = A'\left(\frac{i}{\omega\mu}\right)\left(\frac{\pi}{a}\right)\left(\frac{\pi}{d}\right) sin\left(\frac{\pi}{a}x\right) cos\left(\frac{\pi}{d}z\right)$$

$$H_y = 0$$

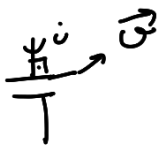
$$H_z = -A'\left(\frac{i}{\omega\mu}\right)\left[\left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{b}\right)^2\right] cos\left(\frac{\pi}{a}x\right) sin\left(\frac{\pi}{d}z\right)$$

where it should be noted that

- the only one electric field component and two magnetic field components are present.
- there is an imaginary "i" present in magnetic field, but not in electric fields. This corresponds to a $\pi/2$ phase difference between $\vec{E}$ and $\vec{H}$ fields as

$$i = e^{i\pi/2} = \underbrace{cos(\pi/2)}_{0} + i\,\underbrace{sin(\pi/2)}_{1} = i$$

$$e^{i\varphi} = e^{i\pi/2} \quad \Rightarrow \quad \varphi = \frac{\pi}{2} \;\; phase\ difference.$$



$$i = C\frac{d}{dt}v$$

$$\hat{I} = iwC\hat{V}$$

$$e^{i\pi/2}$$

A rectangular resonator filled with polyethylene ($\varepsilon_r = 2.25, \mu_r = 1$) material with dimensions of $a = 4.755$ cm, $b = 2.215$ cm and $d = 2.200$ cm is made from a piece of copper ($\sigma_{Cu} = 5.813 \times 10^7$ S/m). The resonance frequency of the resonator is to occur at $f = 5$ GHz for the first (dominant) mode ($n = 1$). Find the quality factor of the resonator for the first mode.

Solution

The quality factor Q due to the lossy cavity walls is

$$Q = \omega \frac{W_{average}}{P_{lost}} = \omega \frac{W_E + W_H}{P_{Wall}}$$

$$\left.\begin{aligned} W_E &= \frac{\varepsilon}{4}\int_v \left|\vec{\boldsymbol{E}}_n(\vec{\boldsymbol{r}})\right|^2 dv = \varepsilon \frac{abd}{16} E_0^2 \\ W_H &= \frac{\mu}{4}\int_v \left|\vec{\boldsymbol{H}}_n(\vec{\boldsymbol{r}})\right|^2 dv = \mu \frac{abd}{16} E_0^2 \frac{\varepsilon}{\mu} \end{aligned}\right\} \Rightarrow \quad Q = 2\omega \frac{W_E}{P_{Wall}}$$

$$P_{Wall} = \frac{R_s}{2}\int_S \left|\vec{\boldsymbol{H}}_t\right|^2 ds = \frac{R_s E_0^2 \lambda^2}{8Z^2}\left(n^2 \frac{ab}{d^2} + \frac{bd}{a^2} + n^2 \frac{a}{2d} + \frac{d}{2a}\right)$$

$$Q = 2\omega \frac{W_E}{P_{Wall}} = 2\omega \frac{\varepsilon \frac{abd}{16} E_0^2}{\frac{R_s E_0^2 \lambda^2}{8Z^2}\left(n^2 \frac{ab}{d^2} + \frac{bd}{a^2} + n^2 \frac{a}{2d} + \frac{d}{2a}\right)} = \frac{k^3 abdZ}{4\pi^2 R_s} \frac{1}{\left(n^2 \frac{ab}{d^2} + \frac{bd}{a^2} + n^2 \frac{a}{2d} + \frac{d}{2a}\right)}$$

where $k = \omega\sqrt{\varepsilon\mu}$, $Z = \sqrt{\mu/\varepsilon}$ and $R_S = \sqrt{\omega\mu_0/2\sigma}$.

Now, let us calculate the parameters

$$k = \omega\sqrt{\varepsilon\mu} = \omega\sqrt{\varepsilon_r \varepsilon_0 \mu_0} = \frac{\omega\sqrt{\varepsilon_r}}{c_0} = \frac{2\pi 5.10^9 \sqrt{2.25}}{3.10^8} = 50\pi = 157.08 \text{ rad/s}$$

$$Z = \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_0}{\varepsilon_r \varepsilon_0}} = \sqrt{\frac{1}{\varepsilon_r}} Z_0 = \sqrt{\frac{1}{2.25}} 120\pi = 251.3\ \Omega$$

$$R_s = \sqrt{\frac{\omega\mu_0}{2\sigma}} = \sqrt{\frac{2\pi 5.10^9 \times 4\pi.10^{-7}}{2 \times 5.813 \times 10^7}} = 1.84.10^{-2}\ \Omega$$

Then, the values of Q for $n = 1$ is

$$Q = \frac{k^3 abdZ}{4\pi^2 R_s} \frac{1}{\left(n^2 \frac{ab}{d^2} + \frac{bd}{a^2} + n^2 \frac{a}{2d} + \frac{d}{2a}\right)} = 8403$$

$$Q = \frac{(157.08)^3 \times 4.755 \times 2.215 \times 2.200 \times 251.3 \times 10^{-6}}{4\pi^2 \times 1.84.10^{-2}} \frac{1}{\left(1^2 \frac{4.755 \times 2.215}{2.200^2} + \frac{2.215 \times 2.200}{4.755^2} + 1^2 \frac{4.755}{2 \times 2.200} + \frac{2.200}{2 \times 4.755}\right)}$$

$$Q = 8403$$

The higher Q is desirable (a, b and d must be in meter unit).

An empty waveguide of a rectangular cross section with dimension $a = 0.75$ cm and $b = 2a$ is used to transmit a wave of frequency $= 15$ GHz .

a) What are the propagating TE modes?
b) Can this waveguide be used to transmit a wave of frequency 7.5 GHz ? Explain why.
c) To transmit the wave mentioned in **b)** through the dominant mode, one fills the waveguide with a nonconducting simple dielectric of constitutive parameters $\varepsilon = 4\varepsilon_0$ and $\mu = \mu_0$. Now, can this waveguide be used to transmit a wave of frequency 7.5 GHz ? Explain why.

Solution

a) For the empty waveguide ($\varepsilon = \varepsilon_0$ and $\mu = \mu_0$), $c = c_0 = 1/\sqrt{\varepsilon_0\mu_0}$

$$f_{cutoff} = f_{mn} = \frac{c_0}{2\pi}k_{mn} = \frac{c_0}{2\pi}\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} = \frac{c_0}{2a}\sqrt{m^2 + \frac{n^2}{4}}$$

$$f_{01} = \frac{c_0}{4a} = \frac{3.10^8}{4 \times 75.10^{-4}} = 10^{10} = 10 \text{ GHz}$$

$$f_{10} = f_{02} = \frac{c_0}{2a} = \frac{3.10^8}{2 \times 75.10^{-4}} = 0.2 \times 10^{10} = 20 \text{ GHz}$$

$$f_{11} = \frac{c_0}{4a}\sqrt{5} = \frac{3.10^8}{4 \times 75.10^{-4}}\sqrt{5} = 10^{10}\sqrt{5} = 10\sqrt{5} \text{ GHz}$$

$$f_{12} = \frac{c_0}{4a}2\sqrt{2} = \frac{3.10^8}{4 \times 75.10^{-4}}2\sqrt{2} = 10^{10}0.2\sqrt{2} = 20\sqrt{2} \text{ GHz}$$

$$f_{03} = \frac{c_0}{4a}3 = \frac{3.10^8}{4 \times 75.10^{-4}}3 = 0.3 \times 10^{10} = 30 \text{ GHz}$$

$$\Rightarrow \quad f_{01} < f = 15 \text{ GHz} < f_{10} = f_{02} < f_{11} < f_{12} < f_{03}$$

Only TE_{01} mode can propagate since $f_{01} < f = 15$ GHz. The only one mode is known as a dominant mode.

b) This waveguide cannot be used to transmit a signal of frequency $f = 7.5$ GHz since $f = 7.5$ GHz frequency is less than the dominant cut-off frequency $f = 10$ GHz.

c) When the waveguide is filled with a material of $\varepsilon = 4\varepsilon_0$ and $\mu = \mu_0$, the propagation velocity is changed to

$$c_0 = \frac{1}{\sqrt{\varepsilon_0\mu_0}} \quad \Rightarrow \quad c = \frac{1}{\sqrt{4\varepsilon_0\mu_0}} = \frac{1}{\sqrt{4}}\underbrace{\frac{1}{\sqrt{\varepsilon_0\mu_0}}}_{c_0} = \frac{c_0}{\sqrt{4}}$$

then, new cut-off frequencies of TE modes are

$$f'_{mn} = \frac{c_0}{2\pi\sqrt{4}}k_{mn} = \frac{c_0}{4\pi}k_{mn} = \frac{c_0}{4\pi}\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} = \frac{c_0}{4a}\sqrt{m^2 + \frac{n^2}{4}}$$

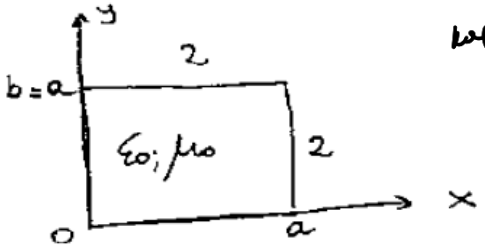
$$\Rightarrow \quad f'_{01} = 5 \text{ GHz} < f' = 7.5 \text{ GHz} < f'_{10} = f'_{02} = 10 \text{ GHz} < f'_{11} = 5\sqrt{5} \text{ GHz} < f'_{12} = 10\sqrt{2} \text{ GHz}.$$

The wave with the frequency of $f' = 7.5$ GHz can now be carried via this waveguide because the new dominant cut-off frequency is less than $f' = 7.5$ GHz . Therefore, this signal can be carried through TE_{01} mode.

An air-filled waveguide of square cross section with the edge length $a = 2$ cm is used to transmit a wave having the wavelength $\lambda = 3$ cm.

a) What are the propagating TE modes? Find the wavelengths of the propagating modes.
b) Can this waveguide be used to transmit a wave of wavelength $\lambda = 6$ cm?
c) To transmit the wave mentioned in **b)** through minimum number of modes, one fills the waveguide with a non-conducting simple dielectric material of constitutive parameters $\varepsilon = p \times \varepsilon_0$ and $\mu = \mu_0$. Determine the number p and the resulting propagating modes.

Solution



$$f_{mn} = \frac{c_0}{2\pi} k_{mn} \quad \Rightarrow \quad k_{mn} = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} = \frac{\pi}{a}\sqrt{m^2+n^2}$$

$$m + n \neq 0 \quad ; \quad m, n = 0,1,2, \ldots$$

a) The condition for propagation

$$k_{mn} < k \quad \Rightarrow \quad \frac{\pi}{a}\sqrt{m^2+n^2} < \frac{2\pi}{\lambda} \quad \Rightarrow \quad \boxed{\sqrt{m^2+n^2} < \frac{2a}{\lambda}.}$$

Let us check the propagation condition for the first two modes

$$\sqrt{m^2+n^2} < \frac{2\times 2}{3} = \frac{4}{3} \quad \Rightarrow \quad \left.\begin{array}{lll} m=0, n=1 & \Rightarrow & 1 < 4/3 \quad k_{01} \\ m=1, n=0 & \Rightarrow & 1 < 4/3 \quad k_{10} \end{array}\right\} \quad \Rightarrow \quad \boxed{k_{01} = k_{10}} \; ; \; \text{two propagating modes.}$$

The propagation constant k_z is the same for the two propagating modes because of $k_{01} = k_{10}$. Therefore, we can consider the calculation for only λ_{01}

$$k_z = \sqrt{k^2 - k_{01}^2} = \sqrt{k^2 - \frac{\pi^2}{a^2}}$$

where $(\lambda_{01} = \lambda_{10})$

$$\lambda_{01} = \frac{v_{01}}{f} = \frac{2\pi}{k_z} \quad \Rightarrow \quad \lambda_{01} = \frac{2\pi}{\sqrt{k^2 - \frac{\pi^2}{a^2}}} = \frac{2\pi}{\sqrt{\left(\frac{2\pi}{\lambda}\right)^2 - \frac{\pi^2}{a^2}}} = \frac{2}{\sqrt{\frac{4}{9}.10^4 - \frac{1}{4}.10^4}}$$

$$\Rightarrow \quad \lambda_{01} = \frac{2.10^{-2}\ \text{m}}{\sqrt{\frac{4}{9} - \frac{1}{4}}} = \frac{2.10^{-2}\ \text{m}}{\frac{\sqrt{16-9}}{6}} = \frac{12\ \text{cm}}{\sqrt{7}} = \boxed{\frac{12\sqrt{7}}{7}\ \text{cm}}$$

b)

$$\left.\begin{array}{l} \lambda = 6\ \text{cm} \quad \Rightarrow \quad k = \dfrac{2\pi}{\lambda} = \dfrac{2\pi}{6} = \dfrac{\pi}{3} \\ \underbrace{k_{01} = k_{10}}_{k_{min}} = \dfrac{\pi}{a}\sqrt{1^2+0^2} = \dfrac{\pi}{a} = \dfrac{\pi}{2} \end{array}\right\} \quad \Rightarrow \quad \frac{\pi}{2} \not< \frac{\pi}{3} \quad \Rightarrow \quad \boxed{k_{min} \not< k}$$

This waveguide cannot be used to transmit a wave of $\lambda = 6$ cm .

c)

$$\underbrace{k_{01} = k_{10}}_{k_{min}} = \frac{\pi}{a} < k_{11} < k_{20} = k_{02} \ldots$$

$$\left.\begin{array}{l} \varepsilon = p \times \varepsilon_0 \\ \mu = \mu_0 \end{array}\right\} \Rightarrow \quad k = \omega\sqrt{\varepsilon\mu} = \omega\sqrt{p\varepsilon_0\mu_0} = \underbrace{\omega\sqrt{\varepsilon_0\mu_0}}_{k_0}\sqrt{p}$$

For minimum number of mode propagation, the condition is ($k = k_0\sqrt{p}$)

$$k_{01} = k_{10} < k < k_{11}$$

$$\Rightarrow \quad k_{01} = k_{10} < k = k_0\sqrt{p} < k_{11} \quad \Rightarrow \quad \frac{\pi}{a} < \underbrace{k_0}_{2\pi/\lambda}\sqrt{p} < \frac{\pi}{a}\sqrt{2}$$

$$\Rightarrow \quad \frac{\pi}{2} < \frac{2\pi}{6}\sqrt{p} < \frac{\pi}{2}\sqrt{2} \quad \Rightarrow \quad \frac{3}{2} < \sqrt{p} < \frac{3}{2}\sqrt{2} \quad \Rightarrow \quad \frac{9}{4} < p < \frac{18}{4} \quad \Rightarrow \quad \boxed{2.25 < p < 4.5}$$

The wavenumbers k_{mn} associated with a rectangular waveguide are ordered according to their greatness as follows:

$$k_1 = k_{10} \;\; ; \;\; k_2 = k_{20} \;\; ; \;\; k_3 = k_{01} \;\; ; \;\; k_4 = k_{30} \;\; ; \;\; k_5 = k_{11}$$

a) What will be the next number k_6?
b) Give the possible numerical value of k_6 if $a = 3b$.

Solution

$$f_{mn} = \frac{c_0}{2\pi} k_{mn} \quad \Rightarrow \quad k_{mn} = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

$$m + n \neq 0 \;\; ; \;\; m, n = 0,1,2, \ldots$$

$$k_1 = k_{10} = \sqrt{\left(\frac{1\pi}{a}\right)^2 + \left(\frac{0\pi}{b}\right)^2} = \frac{\pi}{a}$$

$$k_2 = k_{20} = \sqrt{\left(\frac{2\pi}{a}\right)^2 + \left(\frac{0\pi}{b}\right)^2} = \frac{2\pi}{a}$$

$$k_3 = k_{01} = \sqrt{\left(\frac{0\pi}{a}\right)^2 + \left(\frac{1\pi}{b}\right)^2} = \frac{\pi}{b}$$

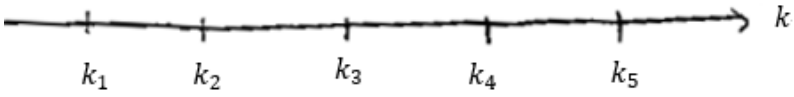
$$k_4 = k_{30} = \sqrt{\left(\frac{3\pi}{a}\right)^2 + \left(\frac{0\pi}{b}\right)^2} = \frac{3\pi}{a}$$

$$k_5 = k_{11} = \sqrt{\left(\frac{1\pi}{a}\right)^2 + \left(\frac{1\pi}{b}\right)^2} = \pi\sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2}$$

a) Finding the next number k_6

$\frac{2\pi}{a} < \frac{\pi}{b} < \frac{3\pi}{a}$

$2b < a < 3b$



$$k_1 < k_2 < k_3 < k_4 < k_5 \quad \Rightarrow \quad \frac{\pi}{a} < \frac{2\pi}{a} < \frac{\pi}{b} < \frac{3\pi}{a} < \pi\sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2}$$

The four possible next numbers are

$$k_{02} = \frac{2\pi}{b} \quad , \quad k_{40} = \frac{4\pi}{a} \quad , \quad k_{12} = \pi\sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{2}{b}\right)^2} \quad , \quad k_{21} = \pi\sqrt{\left(\frac{2}{a}\right)^2 + \left(\frac{1}{b}\right)^2}.$$

k_6 bunlar $k_{40}, k_{02}, k_{12}, k_{21}$ olabilir

- Comparison of $k_{02} = 2\pi/b$ with k_{40}

Assume that $k_{02} > k_{40}$ $\Rightarrow$ k_{02} *cannot be* k_6:

$$k_{02} > k_{40} \quad \Rightarrow \quad \frac{2\pi}{b} > \frac{4\pi}{a} \quad \Rightarrow \quad \frac{1}{b} > \frac{2}{a} \quad \Rightarrow \quad \boxed{2b < a}$$

Varsayım doğru bir olanı ulaşmak

Now, check $2b < a$ over the known (given) relation of $k_2 < k_3$

$$k_2 < k_3 \quad \Rightarrow \quad \frac{2\pi}{a} < \frac{\pi}{b} \quad \Rightarrow \quad 2b < a \quad \Rightarrow \quad k_{40} < k_{02} \quad \Rightarrow \quad k_{02} \text{ cannot be } k_6.$$

• Comparison of k_{12} with k_{40}

Assume that $k_{12} > k_{40} \quad \Rightarrow \quad k_{12}$ *cannot be* k_6:

First, let us use the known (given) relations

$$\left.\begin{array}{l} k_2 < k_3 < k_4 \quad \Rightarrow \quad \dfrac{2\pi}{a} < \dfrac{\pi}{b} < \dfrac{3\pi}{a} \quad \Rightarrow \quad 2b < a < 3b \\ k_4 < k_5 \quad \Rightarrow \quad \dfrac{3\pi}{a} < \pi\sqrt{\left(\dfrac{1}{a}\right)^2 + \left(\dfrac{1}{b}\right)^2} \quad \Rightarrow \quad 2\sqrt{2}b < a \end{array}\right\} \quad \Rightarrow \quad \boxed{2\sqrt{2}b < a < 3b}$$

$$2\sqrt{2}b < a < 3b \quad \Rightarrow \quad 2\sqrt{2} < \frac{a}{b} < 3 \quad \Rightarrow \quad 8 < \left(\frac{a}{b}\right)^2 < 9$$

In this case, k_{12} becomes

$$k_{12} = \pi\sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{2}{b}\right)^2} = \frac{\pi}{a}\sqrt{1 + 4\left(\frac{a}{b}\right)^2} \quad \Rightarrow \quad \frac{\pi}{a}\underbrace{\sqrt{33}}_{\cong 5.7} < k_{12} < \frac{\pi}{a}\underbrace{\sqrt{37}}_{\cong 6}$$

Now, check this condition over the assumed relation $k_{12} > k_{40}$

$$k_{12} > k_{40} \quad \Rightarrow \quad 6\frac{\pi}{a} > k_{12} > 5.7\frac{\pi}{a} > \underbrace{\frac{4\pi}{a}}_{k_{40}} \quad \Rightarrow \quad 6 > k_{12} > 5.7 > 4$$

$$\Rightarrow \quad k_{12} > k_{40} \quad \Rightarrow \quad k_{12} \text{ cannot be } k_6.$$

• Comparison of k_{21} with k_{40}

Assume that $k_{21} > k_{40} \quad \Rightarrow \quad k_{21}$ *cannot be* k_6. Otherwise, k_{21} *can be* k_6

First, considering the previous result

$$8 < \left(\frac{a}{b}\right)^2 < 9.$$

In this case, k_{21} becomes

$$k_{21} = \pi\sqrt{\left(\frac{2}{a}\right)^2 + \left(\frac{1}{b}\right)^2} = \frac{\pi}{a}\sqrt{4 + \left(\frac{a}{b}\right)^2} \quad \Rightarrow \quad \frac{\pi}{a}\sqrt{4+8} < k_{21} < \frac{\pi}{a}\sqrt{4+9} \quad \Rightarrow \quad \frac{\pi}{a}\underbrace{\sqrt{12}}_{\cong 3.5} < k_{21} < \frac{\pi}{a}\underbrace{\sqrt{13}}_{\cong 3.6}$$

Now, check this condition over the assumed relation $k_{21} > k_{40}$

$$k_{21} > k_{40} \quad \Rightarrow \quad 3.6\frac{\pi}{a} > k_{12} > 3.5\frac{\pi}{a} > \underbrace{\frac{4\pi}{a}}_{k_{40}} \quad \Rightarrow \quad 3.6 > k_{12} > 3.5 > 4.$$

This inequality is not satisfied. It means that

$$\Rightarrow \quad k_{21} < k_{40} \quad \Rightarrow \quad k_{21} \text{ can be } k_6.$$

All the calculations show that $\boxed{k_6 = k_{21}}$

b) The possible numerical value of k_6

$$a = 3b \quad \Rightarrow \quad \left(\frac{a}{b}\right)^2 = 9$$

$$k_6 = k_{21} = \frac{\pi}{a}\sqrt{4 + \left(\frac{a}{b}\right)^2} = \frac{\pi}{a}\sqrt{4+9} = \frac{\pi}{a}\sqrt{13}$$

In this case, numerical value of k_5 becomes

$$k_5 = k_{11} = \sqrt{\left(\frac{1\pi}{a}\right)^2 + \left(\frac{1\pi}{b}\right)^2} = \pi\sqrt{1 + \left(\frac{a}{b}\right)^2} = \frac{\pi}{a}\sqrt{10}$$



Then, their sorting is

$$k_1 < k_2 < k_3 < k_4 < k_5 < k_6 \quad \Rightarrow \quad \frac{\pi}{a} < \frac{2\pi}{a} < \frac{\pi}{b} < \frac{3\pi}{a} < \frac{\pi}{a}\sqrt{10} < \frac{\pi}{a}\sqrt{13}.$$

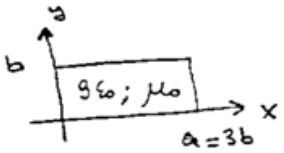
A rectangular waveguide with the ratio of the lengths as $a = 3b$ (a is the length along x dimension) is filled with a non-magnetic and non-conducting dielectric material with the permeability $\varepsilon = 9\varepsilon_0$. The waveguide works at a certain frequency ω_0 at which the TE_{10} and TE_{20} modes propagate.

a) If the waveguide is used at the frequency $2\omega_0$, what will be the propagating TE modes?

b) If the dielectric material is discarded, what will be the propagating TE modes in section **a)**?

c) If the frequency becomes ω_0 in section **b)**, what would be the propagating TE modes?

Solution



ω_0 ; TE_{10} ; TE_{20}

a)

$$f_{mn} = \frac{c_0}{2\pi}k_{mn} \quad \Rightarrow \quad k_{mn} = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} = \frac{\pi}{a}\sqrt{m^2 + 9n^2}$$

$$m + n \neq 0 \quad ; \quad m, n = 0,1,2, \ldots$$

$$k_0 = \omega_0\sqrt{\varepsilon\mu} = \omega_0\sqrt{9\varepsilon_0\mu_0} = 3\omega_0\sqrt{\varepsilon_0\mu_0}$$

$$k_{10} = \frac{\pi}{a}\sqrt{1^2 + 9.0^2} = \frac{\pi}{a}$$
$$k_{20} = \frac{\pi}{a}\sqrt{2^2 + 9.0^2} = \frac{2\pi}{a}$$
$$k_{30} = k_{01} = \frac{\pi}{a}\sqrt{3^2 + 9.0^2} = \frac{3\pi}{a}$$

$$k_{10} < k_{20} < k_0 < k_{30} = k_{01} \quad \Rightarrow \quad \frac{2\pi}{a} < \underbrace{3\omega_0\sqrt{\varepsilon_0\mu_0}}_{k_0} < \frac{3\pi}{a}$$

$$\omega_1 = 2\omega_0 \quad \Rightarrow \quad k_1 = 3\omega_1\sqrt{\varepsilon_0\mu_0} = 6\omega_0\sqrt{\varepsilon_0\mu_0} = 2k_0$$

$$\frac{2\pi}{a} < k_0 < \frac{3\pi}{a} \quad \Rightarrow \quad \frac{4\pi}{a} < \underbrace{2k_0}_{k_1} < \frac{6\pi}{a}$$

$$k_{40} = \frac{\pi}{a}\sqrt{4^2 + 9.0^2} = \frac{4\pi}{a} \qquad k_{11} = \frac{\pi}{a}\sqrt{1^2 + 9.1^2} = \frac{\pi}{a}\sqrt{10} \qquad k_{41} = \frac{\pi}{a}\sqrt{4^2 + 9.1^2} = \frac{\pi}{a}\sqrt{25}$$
$$k_{50} = \frac{\pi}{a}\sqrt{5^2 + 9.0^2} = \frac{5\pi}{a} \qquad k_{21} = \frac{\pi}{a}\sqrt{2^2 + 9.1^2} = \frac{\pi}{a}\sqrt{13} \qquad k_{51} = \frac{\pi}{a}\sqrt{5^2 + 9.1^2} = \frac{\pi}{a}\sqrt{34}$$
$$k_{60} = k_{02} = \frac{\pi}{a}\sqrt{6^2 + 9.0^2} = \frac{6\pi}{a} \qquad k_{31} = \frac{\pi}{a}\sqrt{3^2 + 9.1^2} = \frac{\pi}{a}\sqrt{18} \qquad k_{61} = \frac{\pi}{a}\sqrt{6^2 + 9.1^2} = \frac{\pi}{a}\sqrt{45}$$

The propagating modes are (11 modes)

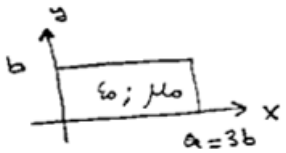
$$k_{01}, k_{10}, k_{20}, k_{30}, k_{40}, k_{50}, k_{11}, k_{21}, k_{31}, k_{41}, k_{51}.$$

In other sense,

$k_{01}, k_{10}, k_{20}, k_{30}, k_{40}, k_{11}, k_{21}$: certainly propagates

$k_{50}, k_{31}, k_{41}, k_{51}$: possibly propagates

b)



$$\omega_2 = 2\omega_0 \quad \Rightarrow \quad k_2 = 2\omega_0\sqrt{\varepsilon_0\mu_0} = \frac{k_1}{3} = \frac{2k_0}{3}$$

$$\frac{4\pi}{3a} < k_2 = \frac{2k_0}{3} < \frac{6\pi}{3a} = \frac{2\pi}{a} \quad \Rightarrow \quad k_{10} = \frac{\pi}{a} < k_2 < k_{20}.$$

The only propagating mode is a dominant mode TE_{10} mode.

c)

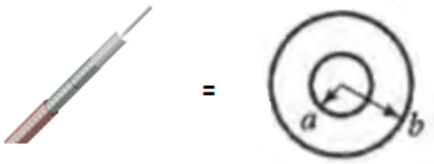
$$\omega_3 = \omega_0 \quad \Rightarrow \quad k_3 = \omega_0\sqrt{\varepsilon_0\mu_0} = \frac{k_0}{3}$$

$$\frac{2\pi}{3a} < k_3 = \frac{k_0}{3} < \frac{\pi}{a} = k_{10}.$$

There is no propagating mode because the wavenumber k_3 is smaller than the dominant mode TE_{10} mode.

An RG-142 type coaxial cable is used for wide-band signal transmission. The inner and the outer radius of this cable is $a = 0.0889$ cm and $b = 0.29464$ cm. A dielectric material with $\varepsilon_r = 2.2$ is used to fill the cable. What is the highest usable frequency before the TE_{11} waveguide mode starts to propagate?

Solution



The cut-off frequency of the TE_{11} mode in the coaxial-cable is

$$f_{cutoff} = f_1 = \frac{c_0}{2\pi\sqrt{\varepsilon_r \mu_r}}\left(\frac{2}{a+b}\right)$$

$$c_0 = 3 \times 10^8 \text{ m/sec}$$
$$\varepsilon_r = 2.2 \quad ; \quad \mu_r = 1$$
$$a = 0.0889 \text{ cm} = 0.000889 \text{ m}$$
$$b = 0.29464 \text{ cm} = 0.0029464 \text{ m}$$

$$f_{cutoff} = \frac{3 \times 10^8 \text{ m/sec}}{2\pi\sqrt{2.2 \times 1}}\left(\frac{2}{0.000889 \text{ m} + 0.0029464 \text{ m}}\right)$$

$$f_{cutoff} = \frac{3 \times 10^8 \text{ m/sec}}{\pi \times 1.483}\left(\frac{1}{0.0038354 \text{ m}}\right)$$

$$f_{cutoff} = \frac{3 \times 10^8}{4.81}\left(\frac{1}{0.0038354}\right) \text{ Hz}$$

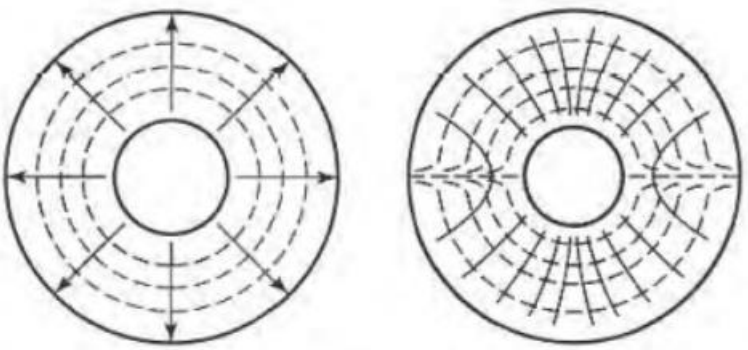
$$f_{cutoff} = \frac{3 \times 10^8}{0.01845} \text{ Hz} = 162.6 \times 10^8 \text{ Hz}$$

$$f_{cutoff} = 16.26 \text{ GHz}$$

In practice, a 5 % safety margin is recommended that

$$f_{cutoff} = 0.95 \times 16.26 \text{ GHz} = 15.45 \text{ GHz}.$$

The field lines of the TEM and TE_{11} modes are shown below.



The field lines of the TEM (left) mode and TE_{11} (right) mode.